

Assurance in Coercive Bargaining

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ABSTRACT. Political actors can fail to secure concessions despite credible threats when targets fear that yielding could invite further harm. We show that this assurance problem can complicate signaling through coercive demands, making it harder for targets to distinguish restraint from plans to attack. Our bargaining model allows concessions to improve the demanding state's military position before it decides whether to attack. This gives states planning an attack a reason to imitate the modest demands of states seeking peace. Greater military strength can make reassuring demand patterns harder to sustain and permit additional ways to conceal an intended attack. These changes can reduce the concessions the demanding state obtains peacefully and increase the risk of war, imposing collective costs when disputes that could have been settled peacefully instead lead to fighting. Lack of assurance can exacerbate information problems in interstate politics, undermining coercive bargaining and making peace harder to secure.

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INTRODUCTION

Political actors use threats to obtain concessions, but targets may resist even credible threats when compliance does not guarantee safety (Schelling 1966). During the Iran nuclear negotiations, Iranian officials feared that an agreement would leave sanctions in place or invite new demands. The Obama administration sought to address these concerns by separating nuclear sanctions from other disputes and managing actors who might obstruct an agreement. Iran accepted the agreement in 2015. In 2018, the Trump administration withdrew and reimposed sanctions to seek further concessions, and Iran subsequently defied the demands and expanded uranium enrichment (Pauly 2024). These events illustrate the assurance problem: persuading the target that compliance will end the pressure. We examine how this assurance problem shapes the demands states make and what targets can learn from them.

We develop a parsimonious model of conflict by relaxing the standard bargaining assumption that compliance ends the crisis. The demanding state knows its own cost of war but the target does not. It makes a demand that the target accepts or resists. If the target resists, the state can fight or back down. If the target complies, the state can keep the settlement or attack, and the concession improves its chance of victory. We compare this setting with an otherwise identical model in which compliance guarantees peace.

Our main result is that the assurance problem itself can make demands harder to interpret, leaving the target less able to distinguish restraint from plans to attack. When concessions improve military prospects, a state planning an attack can imitate one seeking a peaceful settlement to obtain compliance. A modest demand can therefore come from a state that wants to avoid fighting or from one that wants a concession before fighting. In one such pattern, a larger demand can be more reassuring because the state making it is willing to stop if the target complies. Other patterns allow a middle demand to conceal plans to attack even when both smaller and larger demands assure peace.

These changes in what states demand also affect what they obtain peacefully and who bears the costs of war. A state planning an attack can gain by choosing a modest demand that is more likely to be accepted and then attacking after receiving the concession. The target can therefore concede more often while the demanding state obtains less through peaceful settlement. If fear of attack instead leads the target to concede less, the demanding state can lose as a peaceful settlement becomes less valuable. Smaller concessions also make war more attractive to states that would otherwise stop. In both cases, additional wars increase the two states' combined losses.

We contribute to three literatures. First, we provide a parsimonious framework that isolates how assurance affects coercive bargaining. The assurance literature has developed largely through qualitative arguments and case studies. [Pauly \(2024\)](#) explains how separating disputes and managing other actors can help states honor their assurances. [Cebul, Dafoe, and Monteiro \(2021\)](#) use a survey experiment to examine how reputations for restraint affect the perceived credibility of assurances. Following [Paine and Tyson \(2020\)](#), we adopt an experimental approach to formal modeling: we compare otherwise identical games that differ only in whether the coercer can attack after compliance. This framework allows us to trace the effects of lacking assurance through bargaining behavior to concessions, war, and welfare.

Second, we contribute to research on what demands reveal about intentions. [Kydd and McManus \(2017\)](#) show how an aggressive state can imitate the larger demand and assurance of a state willing to settle; the smallest demand remains reassuring in their example. [Slantchev \(2010\)](#) shows that strong states can imitate weak states' modest demands to preserve the element of surprise if refusal leads to war. Acceptance ends the crisis in his model. Here, deception seeks compliance itself: the concession improves the coercer's position for a subsequent attack. This motive can disrupt the ordering of demands as signals of restraint. A smaller demand can be less reassuring than a larger one, and a middle demand can conceal plans to attack even when smaller and larger demands assure peace.

Third, we contribute to research on the limits of military strength in coercive bargaining. Sechser (2010) studies signaling by a target whose war cost is private: resistance can build a reputation for resolve that induces smaller demands in future disputes. Our model reverses these roles, placing private war costs with the coercer and making its demand the signal. When compliance leaves the target exposed to attack, greater strength can enable additional patterns of deception that are unavailable at lower strength. Lack of assurance can also weaken the effect of military strength on coercive ability: increases in strength can yield smaller gains in the average peaceful concession obtained when the target complies.

THE FRAMEWORK

Two states bargain over a disputed good valued at 1. The demanding state, or *coercer*, privately observes its cost of war; the target knows only its distribution. We compare games that differ only in whether the coercer can attack after the target complies. Figure 1 previews the structure of the two games.

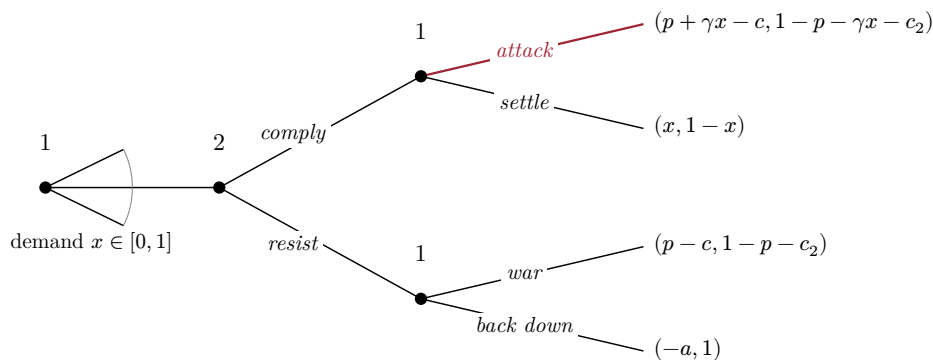


Figure 1: Coercive bargaining games with and without attack after compliance

Note: Player 1 is the coercer and player 2 the target; the coercer's payoff appears first. The tree begins after the coercer learns its war cost c . The target observes the demand but cannot observe this cost, so it cannot distinguish between its decision nodes following the same demand. Under protected compliance, yielding ends the crisis in settlement. Under exposed compliance, the red branch permits attack after yielding, and a concession x raises the coercer's probability of victory from p to $p + \gamma x$.

Actions and payoffs

The coercer chooses how much of the disputed good to demand. Write this demand as $x \in [0, 1]$. The target observes the demand and either complies, transferring that amount, or resists. If the target resists, the coercer chooses between war and backing down.

War is costly for both states. The coercer privately knows its cost, c , which is uniformly distributed on $[0, \bar{c}]$. This privately known cost defines the coercer's *type*. The target's cost, $c_2 \geq 0$, is known to both states. If the coercer's initial probability of victory is $p \in (0, 1)$, war after resistance gives the two states payoffs

$$(p - c, 1 - p - c_2)$$

Backing down imposes an *audience cost*, a political penalty for retreating from a threat, of $a > 0$. The target keeps the good, yielding $(-a, 1)$. A coercer with a sufficiently low war cost fights after resistance; one with a higher cost backs down. We call these types *resolved* and *weak*, respectively.

We compare two games that differ in what happens after compliance. Under the game with *protected compliance*, the transfer ends the crisis, with payoffs $(x, 1 - x)$. Under the game with *exposed compliance*, the coercer can keep this settlement or attack. Receiving the concession first improves its probability of victory by γx , so an attack gives

$$(p + \gamma x - c, 1 - p - \gamma x - c_2) \tag{1}$$

The parameter γ measures how much military advantage a concession provides. It can represent the effect of demobilization, surrendered terrain, or access to strategically valuable territory. At Munich in 1938, Czechoslovakia ceded territory containing fortifications that the agreement required it to leave intact.^[1] An attack puts the entire disputed good at stake again. The concession improves the coercer's chance of winning but is not added separately to its war payoff.

^[1]Munich Agreement, 29 September 1938, [arts. 1–4](#); Wilson to the Secretary of State, 6 October 1938, [FRUS 1938, I, doc. 698](#).

We treat this military advantage as irreversible within the game. Actual agreements can address this danger through phased implementation and military safeguards. The 1979 Egypt–Israel peace treaty paired phased Israeli withdrawal from Sinai with limits on military deployments and provision for outside monitoring.^[2]

Only coercers with sufficiently low war costs prefer attacking to keeping the settlement. We call these types *belligerent* at the demand in question. They are willing to fight after resistance as well as after compliance. A resolved coercer need not be belligerent: it can fight if refused and settle if accepted. Attack after compliance carries no separate reputational penalty in the model. Costly promises and reputations for restraint provide other ways to alter this decision (Kydd and McManus 2017; Cebul, Dafoe, and Monteiro 2021).

We impose two sets of parameter restrictions. The first focuses the analysis on partial concessions while preserving the target’s initial uncertainty about whether the coercer would fight if refused.

ASSUMPTION 1 (Interior bargaining): The parameters satisfy

$$\begin{aligned} p + c_2 &< 1, \quad p + a < \bar{c}, \\ \bar{c} &> 2p + c_2, \quad (p + c_2)(p + a) < p\bar{c}. \end{aligned}$$

The first condition makes the target prefer war after resistance to surrendering the entire good. The second ensures that both resolved and weak coercers are possible. The last two restrict bargaining over a single accepted demand to partial concessions. The third makes the target’s expected gain from accepting this shared demand rather than resisting decrease as the amount demanded rises. The fourth keeps the target’s expected loss from resistance before observing a demand below p . Together, they keep each game’s largest such demand strictly between zero and p .

^[2]Treaty of Peace, 26 March 1979, Annex I, arts. I–II and VI, reproduced by [the American Presidency Project](#).

The second assumption allows concessions to improve military prospects while making settlement more attractive relative to attack. Each unit conceded adds one unit to settlement value but only $\gamma \in (0, 1)$ to the expected return from attack. Larger concessions can therefore encourage restraint even as they improve the coercer's military position. We also require victory to remain uncertain after the largest possible concession.

ASSUMPTION 2 (Concessions and military power): The military effect of concessions satisfies

$$0 < \gamma < 1, \quad p + \gamma < 1.$$

If attacking and keeping the concession give the coercer the same payoff, we assume it keeps the concession and stops. If fighting and backing down after resistance give it the same payoff, we assume it backs down. For any given demand, each equality holds at a single war-cost value, which has zero probability under the continuous distribution. As we show below, different demands can give entire groups of coercers the same payoff.

Kydd and McManus (2017) let the target choose how much to concede after observing the coercer's goal and any threat or assurance. We retain their link between concessions and military power, but let the coercer choose a demand that the target accepts or resists. Allowing the coercer's private war cost to vary continuously lets us identify which types change demands and which abandon settlement when the target concedes less.

Equilibrium and scope

We solve for perfect Bayesian equilibria satisfying the Intuitive Criterion (Cho and Kreps 1987). After observing a demand, the target revises its assessment of the coercer's war cost. Each state chooses the action that gives it the highest expected payoff at every stage, given the other state's behavior and the target's assessment. A coercer with a given war cost chooses one demand without randomizing across demands. The target may sometimes comply and sometimes resist only when these choices give it the same expected payoff. The

Intuitive Criterion limits what the target can infer from a demand that no coercer would make in equilibrium. [Supporting Information D](#) specifies these restrictions and the beliefs that sustain each equilibrium.

In *pooling*, only one demand receives any compliance in equilibrium; other coercers may make demands that the target always resists. In *two-demand bargaining*, two demands for nonzero concessions each attract a group of coercers and are accepted at least sometimes. Each demand must be chosen with nonzero probability, so a demand chosen only at an isolated war-cost value does not qualify. We characterize every equilibrium with these two accepted demands, allowing their sizes and acceptance probabilities to vary. Here, *reassurance* means that compliance ends in settlement with certainty; it describes behavior rather than a separate message.

Protected compliance cannot sustain a third demand meeting these conditions ([Proposition 2](#)). Bargaining with one or two accepted demands therefore covers all demand counts that both games can sustain while producing concessions. We also characterize three-demand bargaining under exposed compliance and allow the number of accepted demands to vary when comparing the best outcomes each game can sustain.

Measuring the outcomes of coercive bargaining

Compliance can produce either settlement or an attack. Obtaining agreement therefore differs from obtaining a peaceful concession. We keep that distinction when measuring the outcomes of coercive bargaining.

DEFINITION 1 (Coercive bargaining outcomes): *Coercive ability* is the average peaceful concession per instance of compliance, counting compliance followed by attack as zero. The *expected peaceful transfer* averages peaceful concessions over all crises, including those in which the target resists. *War risk* is the probability

of fighting, whether after resistance or compliance. *Conflict cost* is the expected sum of both states' war costs. *Total welfare* is the sum of their expected payoffs.

The two concession measures answer different questions. Coercive ability describes what compliance delivers on average; the expected peaceful transfer also accounts for how often compliance occurs. When the target never complies, the peaceful transfer is zero and coercive ability is undefined. [Supporting Information A.2](#) gives the formulas for these measures.

WHAT DEMANDS REVEAL

We begin by examining what the size of a demand reveals about the coercer's willingness to stop after compliance. A small demand can reflect opposing motives. A weak coercer seeks a concession that the target is likely to accept because resistance would force it to back down. A belligerent coercer can also value acceptance when the concession improves its position before an attack. Coercers with intermediate war costs can instead prefer a larger demand, risking resistance but settling if the target complies.

In the protected compliance game, a low demand attracts coercers who place more value on avoiding resistance. Among coercers using the two accepted demands, those with lower war costs choose the larger one and those with higher costs choose the smaller one. We call this ordering *monotone sorting*. This ordering can persist in the exposed compliance game, even if some coercers making the higher demand attack after acceptance. Allowing attack also permits belligerent and weak types to choose the same low demand while coercers with intermediate war costs demand more. We call this *feigned restraint*. These are the only two patterns when each coercer prefers its chosen demand to the alternative, except at the war-cost values that divide the groups. We first establish when each pattern is possible and whether acceptance of either demand can guarantee settlement.

PROPOSITION 1 (Lack of assurance changes what demands can reveal): Under Assumptions 1–2, both games allow pooling in which the target complies at least sometimes.

- In the *protected compliance game*, two demands can distinguish coercers with lower and higher war costs while both groups settle after compliance. Feigned restraint is impossible.
- In the *exposed compliance game*, two demands can distinguish coercers with lower and higher war costs while both groups settle after compliance if and only if $\gamma \leq \frac{c_2}{p+c_2}$. Feigned restraint is always possible.

When both demand groups settle after compliance, the larger demand is $p + c_2$. A higher γ makes the concession more valuable as preparation for an attack. A higher p permits a larger concession, which increases the military advantage obtained before an attack. Greater initial strength can therefore undermine settlement after compliance even when γ is unchanged. In [Figure 2](#), equilibria in which both groups settle exist on and below the curve; feigned restraint remains possible on both sides.

Consider feigned restraint in which each coercer prefers its chosen demand to the other demand, except at the war-cost values that divide the groups. Belligerent types choose the low demand to obtain its higher chance of acceptance and then attack from a stronger position. Weak types choose the same demand to avoid resistance. Coercers with intermediate war costs make the high demand, fight if refused, and settle if accepted. A larger demand is therefore more reassuring in this pattern: acceptance ends the crisis, whereas acceptance of the low demand can precede attack.

The distinction between these motives also matters for what the target learns. Compare this pattern with an equilibrium in the protected compliance game, using the same demands and acceptance probabilities. In the exposed compliance game, belligerent types switch from the high demand to the low one. Their choice becomes indistinguishable from the choice of weak types, so demands reveal less about whether the coercer would fight after

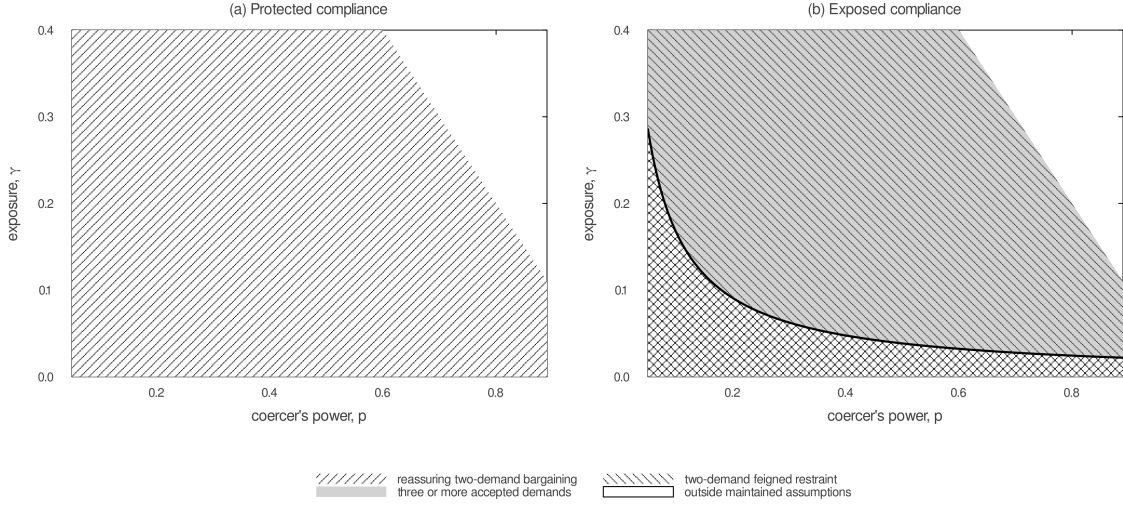


Figure 2: Lack of assurance permits additional demand signals and limits reassurance

Note: Hatching identifies parameter combinations that allow two different demands to receive compliance. In panel (a), two-demand equilibria in which all coercers settle after compliance exist wherever the assumptions hold; feigned restraint and bargaining with three or more accepted demands are impossible. In panel (b), feigned restraint is possible throughout this range of parameters; two-demand equilibria in which all coercers settle after compliance exist on and below the black curve. The gray area strictly above the curve also allows three or more demands to receive compliance. Each demand counted here asks for a nonzero concession, is chosen with nonzero probability, and is accepted at least sometimes. [Proposition 2](#) establishes this existence region for three or more demands; [Supporting Information Y.5](#) gives the proof for larger finite demand counts. Pooling is possible wherever the assumptions hold. The white wedge violates $p + \gamma < 1$. Fixed parameters are $c_2 = 0.02$, $a = 0.30$, and $\bar{c} = 2.00$.

resistance. They do not become less informative about every aspect of war costs: the high demand now identifies coercers with intermediate costs who previously shared their demand with the most belligerent types. [Lemma 5](#) in [Supporting Information W.2](#) establishes this precise information comparison.

This pattern and its outcome comparisons do not depend on war costs being uniformly distributed ([Supporting Information W.1](#)). They remain possible under any continuous distribution that assigns some probability to every interval in the assumed cost range. A different distribution may require different demand sizes or acceptance probabilities. The pooling results and the outcome bounds below continue to assume uniformly distributed war costs.

Signaling through Additional Demands

The assurance problem can also change how many demands bargaining sustains. We next examine whether the target can face more than two accepted demands and what their relative sizes reveal about restraint.

PROPOSITION 2 (Lack of assurance can complicate signaling through demands):

Under Assumptions 1–2, suppose each coercer chooses one demand without randomizing. Consider equilibria with finitely many demands receiving compliance. Each such demand asks for a nonzero concession and is chosen with nonzero probability.

- In the *protected compliance game*, no equilibrium can use three or more such demands.
- In the *exposed compliance game*, an equilibrium with three or more such demands exists if and only if $\gamma > \frac{c_2}{p+c_2}$. When this condition holds, there are equilibria with exactly three demands in which only the middle demand carries a risk of attack after compliance: acceptance of either a smaller or a larger demand always ends in settlement.

The threshold for the military effect of concessions is the same as in [Proposition 1](#).^[3]

In the protected compliance game, every accepted demand above the smallest comes from coercers who would fight if refused and settle if accepted. A larger demand must be accepted less often; otherwise these coercers would all prefer it. To accept each higher demand only sometimes, the target must receive the same expected payoff from accepting it as from resisting. Resistance leads to war after all these demands, while acceptance ends the crisis. Only one concession amount can therefore make acceptance and resistance equally

^[3][Supporting Information X](#) proves that protected compliance permits at most two accepted demands. [Supporting Information Y](#) characterizes all three-demand equilibria under exposed compliance; [Supporting Information Y.5](#) establishes the condition for the finite class with three or more accepted demands. War costs remain uniformly distributed.

attractive. In the exposed compliance game, accepting a higher demand can instead lead to either attack or settlement. The risk of attack can differ across demands, allowing the target to be willing to accept different amounts.

Consider a three-demand equilibrium in which each coercer prefers its chosen demand to both alternatives, except at the war-cost values dividing the groups. Belligerent coercers then use the middle demand to obtain a concession before attacking. They could imitate coercers making the largest demand, but its lower chance of acceptance more than offsets the greater military advantage of receiving it. Given their low war costs, seeking the largest demand and settling if accepted also gives them a lower payoff than making the middle demand and attacking after compliance. Weak coercers use the smallest demand to avoid resistance, while resolved coercers seeking settlement choose among all three. The smallest and largest demands therefore assure peace, while a demand between them can conceal a willingness to attack.^[4]

THE CONSEQUENCES OF ASSURANCE

We now compare the protected and exposed compliance games to assess how assurance affects peaceful concessions, war, and welfare. We begin with pooling and two-demand bargaining, which cover all demand counts that both games can sustain while producing concessions ([Proposition 2](#)). Each form of bargaining can occur in multiple equilibria. We therefore specify how we pair equilibria across the two games.

For the *pooling comparison*, we choose each game’s largest demand that can receive compliance in a pooling equilibrium. The target accepts this demand with the same probability in both games; the demand’s size and the group choosing it can differ.

^[4]The full characterization also covers groups of attackers for whom the low and middle demands give the same payoff, or for whom the middle and highest demands do so. In the latter case, some attackers can choose the highest demand alongside coercers who settle. All three patterns become possible above the same threshold; see [Supporting Information Y](#).

In the *comparison with two demands*, we compare monotone sorting in the protected compliance game with feigned restraint in the exposed compliance game. The low demand has the same size and acceptance probability in both games, as does the high demand. The target always accepts the low demand and accepts the high demand only sometimes. Each coercer prefers its chosen demand to the alternative, except at the war-cost values dividing the groups. Fixing each demand’s size and acceptance probability across games isolates how assurance changes the coercer’s reasons for choosing a demand and attacking after compliance.

For every equilibrium in the exposed compliance game used in this comparison, the proof in [Supporting Information W.1](#) constructs an equilibrium in the protected compliance game with these same demand sizes and acceptance probabilities. In both games, the target obtains the same expected payoff from accepting the high demand as from resisting it. In the protected compliance game, the target strictly prefers accepting the low demand to resisting it. The acceptance probabilities used in the exposed compliance game are therefore also optimal in the protected compliance game. In the exposed compliance game, belligerent coercers switch to the low demand and attack; the other types retain their choices. Overall compliance rises because more coercers choose the demand that is always accepted.

The results therefore describe these specified pairs of equilibria. To examine what happens when bargaining adjusts more broadly, [Proposition 6](#) compares the ranges of possible outcomes while allowing demand sizes, acceptance probabilities, and any finite number of accepted demands for nonzero concessions to vary.

Peaceful Concessions

When coercers share one accepted demand, lack of assurance reduces what the target is willing to concede. In the protected compliance game, the most belligerent types can make a demand the target always resists and then fight. The remaining types make the largest acceptable demand and settle when the target complies. With the attack option, belligerent types gain by making the same demand as coercers seeking settlement: compliance improves

their military position before they fight. Recognizing that some coercers will attack after compliance, the target is willing to accept only a smaller demand.

That defensive response draws additional coercers into war. A smaller concession reduces the value of settlement more than it reduces the return from attacking. Some types that would previously have settled now prefer to fight. War therefore expands for two reasons: concessions directly improve the coercer’s military position, and anticipated exploitation makes the target willing to accept only less valuable peace terms. The second effect reinforces the first.

North Korea voiced this assurance concern in May 2018 when it rejected nuclear disarmament before compensation. Its statement invoked the fate of Libya and Iraq to warn that yielding could leave a state vulnerable to attack (Pauly 2025).^[5]

With two demands, peaceful concessions fall even though compliance becomes more frequent. Belligerent types switch to the low demand, which is accepted more often. But they attack after acceptance. Their high demand would sometimes have produced settlement in the protected compliance game; their low demand now produces no peaceful concession. To capture this loss, we compare both the peaceful concession obtained when the target complies and the peaceful transfer averaged over all crises.

PROPOSITION 3 (Lack of assurance reduces peaceful concessions): Allowing attack after compliance has the following effects:

- In the *pooling comparison*, the expected peaceful transfer and the peaceful concession obtained per instance of compliance both fall.
- In the *comparison with two demands*, both measures fall even though compliance becomes more frequent.

^[5]Kim Kye Gwan’s statement of 16 May 2018, reported in the [Pyongyang Times](#), 17 May 2018, archived by KCNA Watch; contemporaneous quotations appear in Anna Fifield’s [Washington Post](#) report of 16 May 2018.

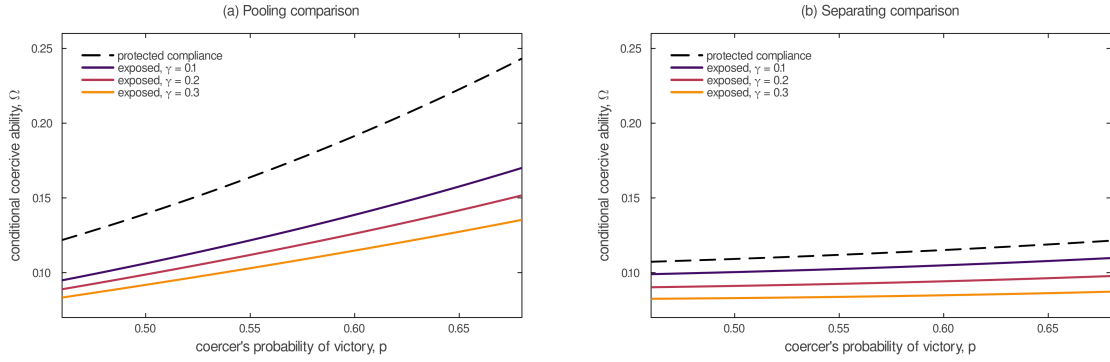


Figure 3: Lack of assurance reduces peaceful concessions

Note: Both panels show peaceful concessions per instance of compliance as the coercer's initial probability of victory increases. The black dashed line represents the protected compliance game. The solid lines represent the exposed compliance game, with the military effect of concessions set to 0.10, 0.20, and 0.30 from dark to light. In panel (a), the target always accepts each game's largest pooling demand; an accepted demand followed by attack counts as no peaceful concession. Panel (b) uses the comparison with two demands: the low demand asks for 10 percent of the good and is always accepted, while the high demand is accepted 8 percent of the time. In both panels, the target's war cost is 0.02, the audience cost is 0.30, and the upper bound on the coercer's war cost is 2.

When concessions provide a greater military advantage, the largest pooling demand the target will accept becomes smaller, and more coercers attack instead of settling. [Supporting Information I](#) gives conditions under which greater initial military strength still increases peaceful concessions, but produces a smaller increase in the exposed compliance game than in the protected compliance game. These are separate changes: initial strength improves the coercer's position before bargaining, while concessions provide an additional military advantage after compliance.

War and Its Costs

The lost settlements raise the probability of war. When coercers share one accepted demand, the additional attackers are those who would have settled if compliance ended the crisis. These coercers fight after resistance in both games. Letting them attack after compliance therefore creates wars out of agreements that would previously have produced settlement.

The same logic applies to the belligerent types that switch demands. In the protected compliance game, their high demand sometimes produces settlement. In the exposed compliance game, they obtain the smaller concession and attack. Their switch removes those peaceful outcomes without changing other types' behavior. To assess the collective cost, we account for both the additional wars and the war costs of the coercers involved.

PROPOSITION 4 (Lack of assurance increases war and conflict costs): Allowing attack after compliance has the following effects:

- In the *pooling comparison*, the probability of war and the expected conflict costs borne by the two states strictly increase.
- In the *comparison with two demands*, both measures strictly increase. The additional wars are exactly the peaceful settlements forgone when belligerent types switch to the low demand and attack.

In the comparison with two demands, the increase in war probability equals the share of belligerent types that switch demands multiplied by the probability that their high demand previously ended in settlement. The increase in expected conflict costs sums the costs of those additional wars. Raising the acceptance probability in both pooling equilibria prevents some wars in each. The figures show the target always accepting each game's largest pooling demand, which produces the lowest war risk among that game's pooling equilibria.

Who Gains and Who Loses

The additional wars impose collective costs, but the coercer may still benefit from being able to attack after compliance. We therefore examine both the total welfare loss and its distribution between the states. In the pooling comparison, weak coercers face the same chance of resistance and the same penalty for backing down in both games. The expected political cost of retreat therefore stays unchanged. At each game's largest pooling demand, accepting and resisting give the target the same expected payoff. Because the same coercers

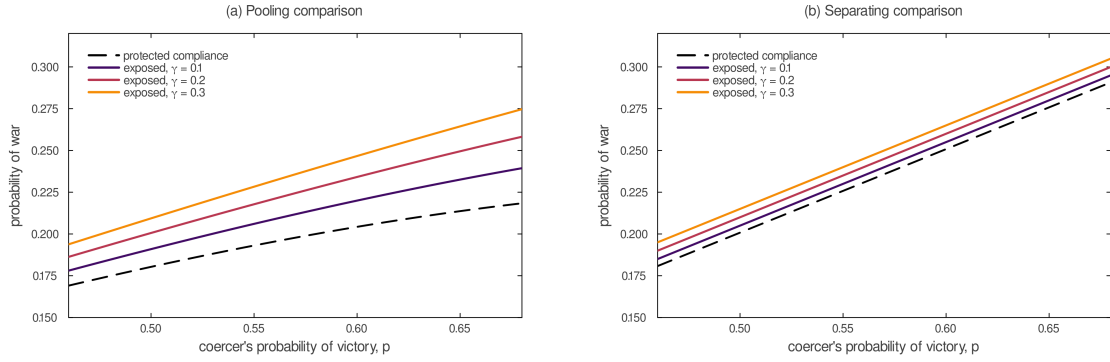


Figure 4: Lack of assurance increases the probability of war

Note: Both panels show the probability of war as the coercer's initial probability of victory increases. Parameters and line meanings are the same as in Figure 3. In panel (a), the target always accepts each game's largest pooling demand. Panel (b) uses the comparison with two demands: belligerent types switch to the low demand and attack after receiving the concession. War includes fighting after either resistance or compliance.

would fight after resistance in either game, the target's expected payoff before observing a demand is also unchanged. The coercer receives less valuable peace terms and loses the full amount of the decline in total welfare.

With two demands, lack of assurance instead benefits the coercer. Because the demand sizes and acceptance probabilities are unchanged, every type can still obtain at least its previous payoff. Belligerent types receive a higher payoff by switching to the more readily accepted demand and then attacking. The target's payoff falls by the coercer's gain plus the additional costs of fighting.

PROPOSITION 5 (Lack of assurance can benefit or harm the coercer while reducing total welfare): Allowing attack after compliance lowers total welfare in both comparisons.

- In the *pooling comparison*, the target’s expected payoff is unchanged and the coercer’s expected payoff falls.
- In the *comparison with two demands*, the coercer’s expected payoff rises, while the target’s payoff falls by the coercer’s gain plus both states’ additional expected war costs.

The pooling loss concerns the coercer’s expected payoff before it learns its war cost; some types may nevertheless gain. The gain with two demands comes from adding the possibility of attack while keeping demand sizes and acceptance probabilities fixed. These comparisons explain why the coercer can gain from the attack option if the target’s willingness to concede is unchanged, yet lose on average if the target will accept only a smaller concession.^[6]

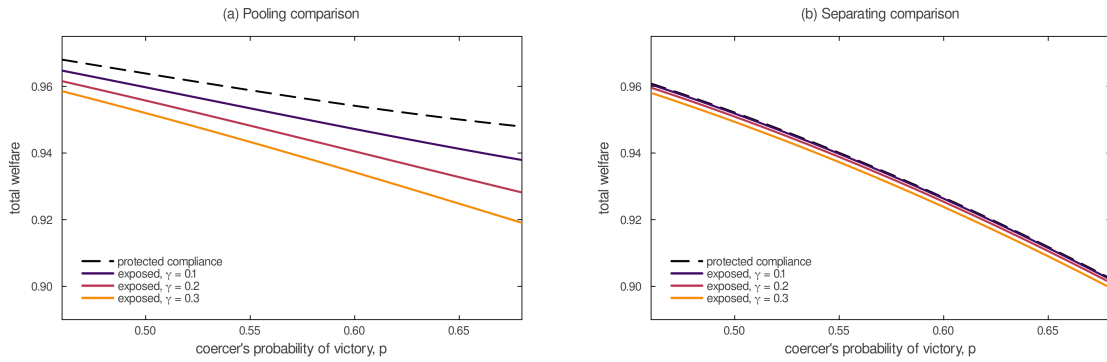


Figure 5: Lack of assurance lowers total welfare

Note: Both panels show the sum of the states’ expected payoffs as the coercer’s initial probability of victory increases. Parameters and line meanings are the same as in Figure 3. In panel (a), the target always accepts each game’s largest pooling demand. Panel (b) uses the comparison with two demands. In both comparisons, the additional wars reduce total welfare.

^[6]Other models show how improved limited-attack capabilities can hurt their holder (Schram 2022) or how switching to a cheaper, weaker campaign can reduce concessions (Spaniel and İdrisoğlu 2024). Their games end when an agreement is accepted; ours adds an attack decision after compliance.

When Bargaining Adjusts

The preceding comparisons fix some features of bargaining across games. We now ask whether bargaining can adjust enough to recover the best outcomes available when compliance guarantees peace. We allow demand sizes, the groups choosing each demand, acceptance probabilities, and the number of accepted demands to vary. War costs remain uniformly distributed, and each coercer chooses one demand without randomizing. There can be any finite number of demands for nonzero concessions, provided each is chosen with nonzero probability and accepted at least sometimes. We also allow a group of coercers to divide between demands when each coercer receives the same payoff from either demand. The comparison thus includes all three-demand equilibria characterized above.

PROPOSITION 6 (Lack of assurance worsens the best attainable outcomes): Under the preceding conditions, allowing attack after compliance has the following effects:

- *War and conflict costs.* The lowest attainable war risk and expected conflict costs rise.
- *Payoffs and welfare.* The highest attainable expected payoff for the coercer and total welfare fall.

In each game, the best outcomes on all four measures occur when the target always accepts the largest pooling demand. Every equilibrium with two or more accepted demands performs strictly worse on all four measures than this pooling equilibrium. Two-demand equilibria can nevertheless approach its outcomes arbitrarily closely.

First suppose the target always accepts the smallest demand. If other demands also receive compliance, some resolved coercers ask for more, risk resistance, and fight if refused. War then involves some coercers with higher costs, rather than only those with the lowest costs. At the same overall probability of war, fighting is therefore more costly on average. The target can refuse any demand, so it accepts only if doing so gives it at least the

expected payoff from resistance. Together, this requirement and the additional conflict costs limit how large the smallest demand can be. It must be below the largest pooling demand, making settlement less valuable to the coercer. Allowing the target to resist even the smallest demand adds another source of war and cannot improve on these pooling outcomes.

The payoffs in this comparison average over all possible war costs. Some coercer types can therefore gain even though the highest attainable average payoff falls. In the equilibrium giving the coercer its highest expected payoff in each game, the target does as well as it would by resisting every demand. The target's expected payoff is therefore unchanged across games, so the additional expected costs of war account for the fall in the coercer's highest expected payoff.

Allowing attack after compliance narrows the range of values each outcome can take. At the other end of each range, both games have the same outcome when the target resists every demand: war risk and conflict costs are highest, while the coercer's payoff and total welfare are lowest. These ranges describe each measure separately; they do not imply that all combinations of values within the ranges are possible. An equilibrium in the exposed compliance game can still perform better on all four measures than a particular equilibrium in the protected compliance game, even when the target always accepts the low demand in each game. [Supporting Information V](#) provides the exact ranges and examples. The result does not cover demands for no concession that the target accepts, infinitely many accepted demands, or coercers who randomize across demands.

EQUILIBRIUM SCOPE

Both games also allow equilibria in which the target resists every demand. Its beliefs about the coercer's war cost can make resistance the best response to every demand, including demands that no coercer makes in equilibrium. No concession then improves the coercer's military position. [Supporting Information H](#) constructs these equilibria.

In [Supporting Information L](#) and [Supporting Information N](#), we characterize all two-demand equilibria under the restrictions on demand choices stated above. In [Supporting Information Y](#), we do so for three demands. These characterizations include cases in which different demands give a coercer the same payoff. The bounds on war, costs, and payoffs also apply when more than three demands receive compliance, provided their number remains finite and they meet the same conditions. We do not fully characterize those larger equilibria.

Comparing pooling equilibria with different probabilities of acceptance can reverse the ordering of outcomes ([Supporting Information J](#)). When concessions provide no military advantage, each game’s largest pooling demand produces the same expected peaceful transfer, war risk, conflict costs, and payoffs if the target accepts it with the same probability. The compliance rate can still differ: some coercers may obtain compliance before fighting in one game and fight after resistance in the other ([Supporting Information W.3](#)).

CONCLUSION

The assurance problem can make demands harder to interpret. A coercer planning an attack can imitate the modest demand of one seeking peace because the concession improves its military prospects. A larger demand can then be more reassuring. Lack of assurance also permits additional demands, including patterns in which a middle demand carries an attack risk while both smaller and larger demands assure peace.

Who gains depends on the target’s response. In pooling, fear of attack leads the target to accept only a smaller demand at the same acceptance probability. Settlement becomes less attractive, and the coercer’s expected payoff falls. In the comparison with two demands, each demand keeps its size and acceptance probability. Belligerent coercers then gain by switching to the low demand, which is always accepted, and attacking after receiving the concession. The target’s loss equals their gain plus the added costs of war. Both comparisons produce fewer peaceful concessions and lower total welfare. The best outcomes also worsen under our broader comparison, which allows demand sizes, acceptance probabilities, and

the number of accepted demands to change. The lowest achievable war risk and conflict cost rise, while the highest coercer payoff and total welfare fall. Particular equilibria in the exposed compliance game can nevertheless produce better outcomes than particular equilibria in the protected compliance game.

Testing feigned restraint requires evidence that belligerent coercers choose the same modest demand as coercers seeking to avoid resistance. It also requires comparison with coercers who ask for more and stop after acceptance. An attack after a concession alone cannot establish this pattern.

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SUPPORTING INFORMATION

CONTENTS

A. Notation and Exact Statements	4
A.1. Parameters and continuation decisions	4
A.2. Outcome measures	5
A.3. Comparing demand choices	5
A.4. Exact forms of the main propositions	6
B. Pooling Equilibrium Formulas	8
C. Preliminary Results	10
D. The Intuitive Criterion and Off-Path Beliefs	11
E. Protected-Compliance Pooling Construction	13
F. Imitation under Positive Compliance	14
G. Exposed-Compliance Pooling Construction	14
H. Equilibria in Which Every Demand Is Resisted	15
I. Pooling Outcome Comparisons	16
J. Independent Compliance Rates at Maximal Pooling Demands	18
K. Two-Demand Sorting Equilibria	19
L. Monotone Two-Pool Sorting	20
M. Proof of the Monotone Characterization	21
M.1. Proof of Proposition 7	21
N. Low Demands Can Feign Restraint Only under Exposed Compliance	23

O.	Proof of the Feigned-Restraint Characterization	26
O.1.	Proof of Proposition 8	26
P.	An Interior Feigned-Restraint Example	27
Q.	Zero-Demand Sorting Boundaries	28
R.	Proof of Equilibrium-Class Existence	29
R.1.	Proof of Proposition 1	29
S.	Two-Pool Outcome Formulas	31
T.	Proofs of the Main Outcome Propositions	32
T.1.	Proof of Proposition 3 through Proposition 5	32
U.	Pure Compliance as a Special Case	34
V.	Bargaining Adjustment and Attainable Outcomes	35
V.1.	Proof of Proposition 6	36
V.2.	Pairwise Reversals with Certain Low-Demand Acceptance	39
W.	Distribution, Information, and Zero-Exposure Benchmarks	40
W.1.	Regular Sorting under a General Prior	40
W.2.	What Demand Signals Reveal	43
W.3.	Retaining the Attack Option at Zero Exposure	44
X.	Accepted Demand Counts across the Two Games	46
Y.	Three-Demand Equilibria under Exposed Compliance	48
Y.1.	Attackers Strictly Prefer the Middle Demand	49
Y.2.	Attackers Are Indifferent between the Low and Middle Demands	50

Y.3. Attackers Are Indifferent between the Middle and High Demands	50
Y.4. Proof of the three-demand claims in Proposition 2	51
Y.5. Higher Demand Counts in the Existence Map	56
Y.6. Outcomes and Comparisons across Demand Counts	57

A. NOTATION AND EXACT STATEMENTS

This section collects the notation, parameter restrictions, and quantitative statements behind the main text. The model and results use the same assumptions here and in the main text; only their presentation differs.

A.1. Parameters and continuation decisions

The coercer's war cost has cumulative distribution function F on $[0, \bar{c}]$, uniform unless an extension states otherwise. Let $h = p + c_2$ denote the target's loss from ordinary war, and use P and E for protected and exposed compliance. Assumption 1 imposes

$$h < 1, \quad r < \bar{c}, \quad \bar{c} > 2p + c_2, \quad hr < p\bar{c}$$

where the resistance cutoff is

$$c < r \equiv p + a \tag{2}$$

The coercer fights after resistance exactly below that cutoff. Assumption 2 imposes $0 < \gamma < 1$ and $p + \gamma < 1$. After exposed compliance with demand x , the coercer attacks exactly when

$$c < t_E(x) \equiv p - (1 - \gamma)x \tag{3}$$

At a continuation tie it chooses the peaceful outcome. The shorthand $t_P(x) = p - x$ identifies the protected type indifferent between settlement at x and ordinary war.

Let $\lambda(x)$ denote the target's probability of compliance after demand x . For a pooling equilibrium, write this probability as λ_j in game $j \in \{P, E\}$. At either game's maximal pooling demand the target is indifferent, so every $\lambda_j \in (0, 1]$ is supportable. The pooling comparison imposes $\lambda_P = \lambda_E = \lambda > 0$. This matches a response to a demand, not the unconditional frequency of compliance.

After an unused demand, the Intuitive Criterion excludes types that cannot attain their equilibrium payoff under any sequentially rational target response, provided some type remains. [Lemma 3](#) supplies supporting beliefs and responses for every construction without changing on-path behavior.

A.2. Outcome measures

Let C denote compliance, S settlement after compliance, W either ordinary war or post-compliance attack, and D the on-path demand. The measures in Definition 1 are

$$\Omega = \frac{E[D\mathbb{1}[S]]}{\Pr(C)}, \quad T = E[D\mathbb{1}[S]], \quad q = \Pr(W)$$

where Ω is defined only for $\Pr(C) > 0$. Expected conflict cost is

$$K = E[(c + c_2)\mathbb{1}[W]]$$

Write $I(z)$ for the expected conflict cost contributed by types below z . The coercer's expected payoff is U , the target's is U_2 , and total welfare is $V = U + U_2$. Conflict cost excludes the audience cost of backing down; total welfare includes it.

A.3. Comparing demand choices

Write x_P and x_E for the largest accepted pooling demands, and $t_P = t_P(x_P)$ and $t_E = t_E(x_E)$ for their associated war cutoffs. Their difference is

$$t_E - t_P = \underbrace{\gamma x_E}_{\text{direct military shift}} + \underbrace{(x_P - x_E)}_{\text{demand feedback}} > 0 \quad (4)$$

The first term is the concession's direct military benefit. The second is the loss in settlement value when the target reduces its concession.

The comparison with two demands matches a regular exposed feigned-restraint equilibrium to protected monotone sorting at the same demands and target responses. Write the demands as $0 < x < y = h$, with certain compliance after x and compliance probability $\rho \in (0, 1)$ after y . Under protection, types below k choose the high demand. Under exposure, types below b switch to the low demand and attack, types in $[b, k)$ retain the high demand and settle if accepted, and types above k retain the low demand. Let A denote attack after compliance. Then

$$\Pr(A \mid x, C) = \frac{b}{b + \bar{c} - k}$$

With Δ denoting exposed minus protected outcomes and P_C denoting the unconditional probability of compliance,

$$\Delta P_C = \frac{(1-\rho)b}{\bar{c}} > 0, \quad \Delta T = -\frac{\rho hb}{\bar{c}} < 0 \quad (5)$$

The proofs below give the incentive and acceptance conditions for this comparison. For pooling, the power comparative static in the main text holds at fixed other primitives over $p < \bar{p} = \frac{\bar{c}-c_2-a}{2}$.

A.4. Exact forms of the main propositions

The following mathematical statements accompany the verbal propositions in the main text. All impose the common model assumptions and equilibrium concept. They preserve the quantitative comparisons used in the proofs.

Exact statement of Proposition 1.

Under Assumptions 1–2:

1. *Pooling.* Both games admit positive-compliance pooling throughout the parameter space.
2. *Separating equilibria.* With two strictly positive demands, protected compliance admits reassuring monotone sorting and excludes feigned restraint. Exposed compliance admits feigned restraint throughout, and reassuring monotone sorting if and only if $\gamma \leq \frac{c_2}{h}$.

Exact statement of Proposition 2.

Under Assumptions 1–2 and pure demand strategies, no PBE in the protected compliance game contains three or more strictly positive on-path demands each chosen by positive prior mass and accepted with positive probability. In the exposed game, an equilibrium satisfying the Intuitive Criterion whose on-path accepted demands form a finite set of at least three strictly positive amounts, each chosen by positive prior mass, exists if and only if

$$\gamma(p + c_2) > c_2.$$

Whenever this condition holds, an equilibrium whose entire set of used demands consists of exactly three such amounts also exists. The strict first family in [Supporting Information Y](#) has $0 < x < z < y = h$: every type choosing x or y settles after compliance, whereas a positive mass choosing z attacks. Demand choices are strict except at cutoff types. Thus the probability of attack conditional on compliance is zero after both extreme demands and strictly positive after the middle demand. [Lemma 7](#) in [Supporting Information X](#) proves the protected impossibility.

Supporting Information Y characterizes all three-demand profiles, and Supporting Information Y.5 establishes necessity of the condition for the full finite class. This result concerns the possible demand signals and their relationship to restraint; it does not supply an information ordering across arbitrary equilibria.

Exact statement of Proposition 3.

1. *Pooling.* At the maximal pooling demands and every common compliance rate $\lambda > 0$, $\Omega_E < \Omega_P$ and $T_E(\lambda) < T_P(\lambda)$.
 1. *Separating equilibria.* Matching regular exposed feigned restraint to protected monotone sorting at the same demands and target responses, with certain low-demand acceptance, $\Omega_E < \Omega_P$ and $T_E < T_P$.

Exact statement of Proposition 4.

1. *Pooling.* At the maximal pooling demands and every common compliance rate $\lambda > 0$, $q_E(\lambda) > q_P(\lambda)$ and $K_E(\lambda) > K_P(\lambda)$.
 1. *Separating equilibria.* In the matched comparison, exposure raises war risk by $\frac{\rho b}{\bar{c}}$ and expected conflict cost by $\rho I(b)$.

Exact statement of Proposition 5.

1. *Pooling.* At the maximal pooling demands and every common compliance rate $\lambda > 0$, $V_E(\lambda) < V_P(\lambda)$. The target's expected payoff is unchanged, so the coercer bears the welfare loss: $U_E(\lambda) < U_P(\lambda)$.
 1. *Separating equilibria.* In the matched comparison, total welfare falls by $\rho I(b)$. The coercer gains $\frac{\rho b^2}{2\bar{c}}$, but the target loses $\frac{\rho b(b+c_2)}{\bar{c}}$.

Exact statement of Proposition 6.

Under uniform costs and pure demand strategies, maximal pooling with certain compliance minimizes war risk and conflict costs and maximizes the coercer's expected payoff and total welfare across equilibria whose on-path accepted demands form a finite set of strictly positive amounts, each chosen by positive prior mass. The bounds are strict whenever at least two such demands are used, and two-demand equilibria approach the bounds arbitrarily closely. Exposure strictly raises both minima and lowers both maxima. Accepted zero demands are excluded.

B. POOLING EQUILIBRIUM FORMULAS

This section collects the algebra behind the pooling results in the main text. A pooling equilibrium has one on-path demand receiving positive compliance; other on-path demands, if any, are surely resisted. Responses to unused demands do not enter this classification. The maximal demands below are maximal within this class. They need not bound the demands in separating equilibria. After resistance, a type's continuation value is

$$R(c) = \max\{p - c, -a\}$$

Under protected compliance, the target accepts a demand $x < p$ when

$$x \leq \frac{h(F(r) - F(t_P(x)))}{1 - F(t_P(x))} \quad (6)$$

where $t_P(x) = p - x$. The maximal accepted demand x_P is the positive root of

$$x^2 + (\bar{c} - 2p - c_2)x - ha = 0 \quad (7)$$

Writing $t_P = p - x_P$, the protected outcomes under pure compliance with the accepted demand are

$$\Omega_P = x_P, \quad T_P = x_P(1 - F(t_P)), \quad q_P = F(t_P) \quad (8)$$

Throughout this section, a pooling outcome written without a compliance-probability argument denotes its pure-compliance endpoint. Conditional coercive ability Ω_j is independent of that probability. For later use, define the expected conflict cost contributed by types below $z \in [0, \bar{c}]$ as

$$I(z) \equiv \int_0^z (c + c_2) dF(c) = \frac{z^2 + 2c_2 z}{2\bar{c}}$$

Thus $K_P = I(t_P)$. At x_P , any $\lambda_P \in (0, 1]$ is supportable, giving

$$\begin{aligned} T_P(\lambda_P) &= \lambda_P x_P (1 - F(t_P)) \\ q_P(\lambda_P) &= F(t_P) + (1 - \lambda_P)(F(r) - F(t_P)) \\ K_P(\lambda_P) &= I(t_P) + (1 - \lambda_P)(I(r) - I(t_P)) \end{aligned}$$

Under exposed compliance, the target's indifference condition for an all-type pool is

$$x(1 - F(t_E(x))) + (h + \gamma x)F(t_E(x)) = hF(r) \quad (9)$$

and the maximal accepted demand x_E is the positive root of

$$(1 - \gamma)^2 x^2 + [\bar{c} - (1 - \gamma)(2p + c_2)]x - ha = 0 \quad (10)$$

Let

$$D_E = 2(1 - \gamma)^2 x_E + \bar{c} - (1 - \gamma)(2p + c_2)$$

$$D_P = 2x_P + \bar{c} - 2p - c_2$$

At the maximal demand, coercive ability and the expected peaceful transfer are

$$\Omega_E = x_E(1 - F(t_E)), \quad T_E(\lambda) = \lambda x_E(1 - F(t_E)) \quad (11)$$

War risk and expected conflict costs are

$$q_E(\lambda) = F(t_E) + (1 - \lambda)(F(r) - F(t_E)) \quad (12)$$

$$K_E(\lambda) = I(t_E) + (1 - \lambda)(I(r) - I(t_E)) \quad (13)$$

Implicit differentiation of the exposed demand equation gives

$$\frac{\partial x_E}{\partial \gamma} = -\frac{x_E[2p + c_2 - 2(1 - \gamma)x_E]}{D_E} < 0 \quad (14)$$

$$\frac{\partial t_E}{\partial \gamma} = x_E - (1 - \gamma)\frac{\partial x_E}{\partial \gamma} > x_E \quad (15)$$

The power derivatives of the two maximal demands are

$$\frac{\partial x_P}{\partial p} = \frac{2x_P + a}{D_P}, \quad \frac{\partial x_E}{\partial p} = \frac{2(1 - \gamma)x_E + a}{D_E} \quad (16)$$

Direct integration gives the coercer's expected payoffs at the maximal pooling demands:

$$U_P = x_P + \frac{t_P^2}{2\bar{c}} \quad (17)$$

$$U_P(\lambda_P) = \lambda_P \left[x_P + \frac{t_P^2}{2\bar{c}} \right] + (1 - \lambda_P) \left[-a + \frac{r^2}{2\bar{c}} \right]$$

$$U_E(\lambda) = \lambda \left[x_E + \frac{t_E^2}{2\bar{c}} \right] + (1 - \lambda) \left[-a + \frac{r^2}{2\bar{c}} \right] \quad (18)$$

For either $j \in \{P, E\}$, the target's expected payoff and total welfare are

$$U_{2,j}(\lambda_j) = 1 - hF(r), \quad V_j(\lambda_j) = 1 - K_j(\lambda_j) - (1 - \lambda_j)a[1 - F(r)]$$

In particular, $V_P = 1 - I(t_P)$ and $V_E = 1 - I(t_E)$ at pure compliance. The main pooling comparison sets $\lambda_P = \lambda_E = \lambda > 0$.

C. PRELIMINARY RESULTS

We begin with the coercer's choice between fighting and stopping after the target responds. These decisions determine which war costs lead to fighting after resistance and which lead to an attack after compliance.

LEMMA 1 (Continuation cutoffs): With the stated peaceful tie convention, the coercer fights after resistance exactly when $c < r = p + a$. Under exposed compliance, after compliance with demand x , it attacks exactly when $c < t_E(x) = p - (1 - \gamma)x$.

Proof. War after resistance yields $p - c$, while backing down yields $-a$. Attack after compliance yields $p + \gamma x - c$, while settlement yields x . Comparing the relevant payoffs gives both cutoffs. $\square$

The pooling comparisons require each game to admit a largest accepted demand that is positive and falls short of the entire good. We next verify that the parameter restrictions ensure this and establish how the two demands and their associated attack cutoffs are ordered.

LEMMA 2 (Interior accepted demands): Under Assumptions 1–2, x_P and x_E are the unique positive roots of (7) and (10). They satisfy $0 < x_E < x_P < p$ and $0 < t_P < t_E < r < \bar{c}$.

Proof. Under uniformity, the target's benchmark acceptance surplus, multiplied by $\bar{c}$, is

$$h(a + x) - x(\bar{c} - p + x) = ha - (\bar{c} - 2p - c_2)x - x^2$$

Assumption 1 makes this expression positive at zero, negative at p , and strictly decreasing on $[0, p]$. It therefore has the unique root x_P .

For $0 \leq x \leq p$, exposed compliance gives resistance loss minus compliance loss, multiplied by $\bar{c}$, of

$$h(p + a) - (h + \gamma x)[p - (1 - \gamma)x] - x[\bar{c} - p + (1 - \gamma)x]$$

Setting this expression to zero gives (10). Its constant term is negative and its leading coefficient is positive, so it has exactly one positive root. Evaluating the target's acceptance surplus at $x = p$ gives a negative value, which places x_E in $(0, p)$.

Evaluate the exposed-compliance polynomial at x_P . Using (7), its value is

$$\gamma x_P[2p + c_2 - (2 - \gamma)x_P] > 0$$

The last inequality follows from $x_P < p$. Since the polynomial is negative at zero, $x_E < x_P$. Thus

$$t_E - t_P = \gamma x_E + x_P - x_E > 0$$

The remaining inequalities follow from Assumptions 1–2. □

D. THE INTUITIVE CRITERION AND OFF-PATH BELIEFS

Let $v_j(c)$ denote type c 's equilibrium payoff in game $j \in \{P, E\}$. Its payoff from demand d when the target complies with probability λ is

$$u_P(d, c, \lambda) = (1 - \lambda)R(c) + \lambda d,$$

$$u_E(d, c, \lambda) = (1 - \lambda)R(c) + \lambda \max\{d, p + \gamma d - c\}.$$

Continuation play follows Lemma 1. Let $\Lambda_j(d)$ be the set of compliance probabilities that are optimal for the target under some posterior on $[0, \bar{c}]$. After an unused demand d , the types not ruled out by equilibrium dominance are

$$\mathcal{T}_j(d) = \left\{ c \in [0, \bar{c}] : \max_{\lambda \in \Lambda_j(d)} u_j(d, c, \lambda) \geq v_j(c) \right\}.$$

The Intuitive Criterion rejects an equilibrium outcome if some type in $\mathcal{T}_j(d)$ strictly gains from d under every target best response to beliefs supported on $\mathcal{T}_j(d)$ (Cho and Kreps 1987). If this

set is empty, the criterion imposes no restriction at d . The following construction supplies one admissible belief and response that deters every type at each unused demand.

LEMMA 3 (Completion under the Intuitive Criterion): Under Assumptions 1–2, an on-path profile with pure demand strategies extends to a PBE satisfying the Intuitive Criterion if its posteriors obey Bayes' rule, its target responses are optimal, no type prefers another used demand, and $v_j(c) \geq R(c)$ for every type. The completion leaves all on-path demands, responses, and payoffs unchanged.

Proof. Fix a game and suppress its subscript. Because $r < \bar{c}$, weak types exist. Choose a demand s used by a weak type and let λ_s be its compliance probability. All weak types face identical payoffs from each demand, so sender optimality gives them the common equilibrium payoff

$$w = \lambda_s s - (1 - \lambda_s)a \leq \lambda_s s.$$

Every type can imitate this demand. Its resistance payoff is at least $-a$, and its compliance payoff is at least s in either game. Consequently, $v(c) \geq w$. Under exposure, a resolved type can also attack after imitating s , so

$$v(c) \geq R(c) + \lambda_s \gamma s \quad \text{for } c < r.$$

These bounds, together with $v(c) \geq R(c)$, suffice to specify all unused demands.

First consider $0 \leq d \leq h$ with $d \geq w$. Compliance is a target best response to belief on a resolved type that would settle after compliance. Such a type exists in both games: under exposure, $\max\{0, t_E(d)\} < r$. Thus acceptance is among the responses used to test equilibrium dominance. It gives every weak type $d \geq w$, so weak types belong to $\mathcal{T}(d)$. Put posterior probability one on a type $c > r$ and resist. The target obtains 1 from resistance and $1 - d$ from compliance, making resistance strictly optimal for $d > 0$ and weakly optimal at zero. Every sender then receives $R(c) \leq v(c)$.

Next consider $0 \leq d < w$. Settlement after compliance gives $d < w \leq v(c)$. If exposure instead induces attack, then $c < t_E(d) \leq p < r$, and the compliance payoff is

$$p + \gamma d - c = R(c) + \gamma d < R(c) + \lambda_s \gamma s \leq v(c).$$

Resistance also gives at most $v(c)$. Hence no pure or mixed target response makes the deviation profitable. If $\mathcal{T}(d)$ is nonempty, put belief on a surviving type and choose a target best response. If it is empty, use a weak-type belief and resist. Either choice is sequentially rational and deters every sender. This step can require compliance with an unused demand, but that demand remains unchosen because no type gains from it.

Finally, let $d > h$. Against a resolved settler, compliance costs $d > h$ rather than the war loss h . Against a weak type, it costs $d > 0$ rather than zero. Against an exposed attacker, it adds $\gamma d > 0$ to the war loss. Resistance is therefore strictly optimal under every posterior, so $\Lambda_j(d) = \{0\}$. The surviving types, if any, satisfy $v(c) = R(c)$. Put belief on any such type, or use any posterior if no type survives, and resist. Again, no sender gains.

The cases exhaust $[0, 1]$, including zero demands, payoff ties, and arbitrary on-path compliance probabilities. The prescribed beliefs meet the criterion whenever it restricts beliefs, and target responses deter all deviations. Bayes' rule and optimality on path are unchanged, while [Lemma 1](#) supplies optimal continuation play after either target action. This completes the equilibrium. $\square$

E. PROTECTED-COMPLIANCE POOLING CONSTRUCTION

Proof. We construct the full maximal-demand family. Let types $c < t_P$ demand 1 and types $c \geq t_P$ demand x_P . At demand 1, Bayes' rule places probability one on types that fight after resistance. Because $h < 1$ under Assumption 1, the target resists. At x_P , the posterior is the upper tail $c \geq t_P$. Condition (6) binds, so any compliance probability $\lambda_P \in (0, 1]$ is optimal.

Types below t_P receive $R(c) = p - c > x_P$ and lose $\lambda_P[R(c) - x_P]$ by mimicking. Types above t_P gain $\lambda_P[x_P - R(c)] > 0$ from their accepted demand relative to resistance; the cutoff type is indifferent. Thus on-path incentives hold and every type obtains at least $R(c)$. Apply [Lemma 3](#) with $s = x_P$ and $w = \lambda_P x_P - (1 - \lambda_P)a$ to complete the equilibrium.

The supporting beliefs can be explicit here. At an unused $0 \leq d < w$, put belief on $c = 0$ and comply. This type remains eligible because its equilibrium payoff is $R(0)$, and compliance is optimal since $d < w \leq x_P < h$. Every sender receives less than its equilibrium payoff from this accepted deviation. At an unused $w \leq d \leq h$, put belief on a weak type and resist. At $d >$

h , put belief on $c = 0$ and resist. If $w \leq 0$, the first case is empty and weak-type beliefs support refusal at zero. These choices satisfy the Intuitive Criterion for every $\lambda_P \in (0, 1]$.

To prove maximality within the one-accepted-demand class, suppose its accepted demand were at least p . Every type would weakly prefer it to resistance, strictly so except possibly at the zero-cost endpoint, so its posterior would be the prior. Acceptance would require $x \leq hF(r) = \frac{hr}{c} < p$, a contradiction to Assumption 1. For an accepted $x < p$, the types with $R(c) < x$ strictly prefer its pool to any resisted demand, while types with $R(c) > x$ prefer resistance. Its posterior is therefore the upper tail $c \geq p - x$, up to the indifferent cutoff. Condition (6) implies $x \leq x_P$. $\square$

F. IMITATION UNDER POSITIVE COMPLIANCE

Proof. Fix a positive demand $x > 0$ that receives compliance with probability $\lambda > 0$ and a type that fights after resistance and attacks after compliance. A surely resisted signal yields $p - c$. Mimicking yields

$$\lambda(p + \gamma x - c) + (1 - \lambda)(p - c) = p - c + \lambda\gamma x > p - c$$

Under Assumption 2, $\lambda\gamma x > 0$, so the type cannot be assigned to the surely resisted signal. The argument uses no restriction on the posterior or on whether the target mixes. A zero demand does not yield the strict gain. $\square$

G. EXPOSED-COMPLIANCE POOLING CONSTRUCTION

Proof. Every type strictly prefers a positive demand receiving positive compliance to a surely resisted signal: attackers gain γx , resolved settlers have $x > p - c$, and weak types have $x > -a$ upon compliance. Thus a pooling equilibrium with one positive accepted demand must place every type there. Compliance costs the target at least x , since an attacking type imposes $h + \gamma x = x + t_E(x) + c_2 > x$, while a settling type imposes x . An accepted all-type pool therefore requires $x \leq hF(r) < p$ under Assumption 1.

Let every type demand $x \in (0, x_E]$. The target's resistance loss minus compliance loss is strictly positive for $x < x_E$ and zero at $x = x_E$. It therefore complies purely below x_E and may

comply with any $\lambda \in (0, 1]$ at x_E . The exposed polynomial is strictly increasing on $[0, p]$, so compliance is strictly suboptimal for $x \in (x_E, p)$. The preceding bound excludes $x \geq p$. Thus no positive-compliance pooling equilibrium exists above x_E .

For any demand in this set, a type $c < t_E(x)$ attacks after compliance. Its on-path payoff is $p - c + \lambda\gamma x$, strictly above the $p - c$ from a refused positive deviation. A type $c \in [t_E(x), r)$ settles after compliance and fights after resistance. Because $c \geq p - (1 - \gamma)x$ and $\gamma x > 0$, it has $x > p - c$ and strictly benefits from every positive compliance probability. A type $c \geq r$ obtains $\lambda x + (1 - \lambda)(-a) > -a$. Thus all types prefer the pool to a refused deviation. Below x_E , set $\lambda = 1$; at x_E , use any $\lambda \in (0, 1]$.

All types obtain more than $R(c)$, and on-path sender and target optimality have been verified. Apply Lemma 3 with $s = x$ and $w = \lambda x - (1 - \lambda)a$. In this positive-demand family, weak-type beliefs and resistance work after every unused demand, including zero. For $0 \leq d < w$ and $d > h$, every type is equilibrium dominated, so the criterion imposes no further restriction there. For $w \leq d \leq h$, weak types survive the test and make resistance optimal. Thus the full family satisfies the Intuitive Criterion.

The boundary $x = 0$ also supports an all-type pool. The target strictly prefers compliance because its surplus is $h[F(r) - F(p)] > 0$. Types $c \leq p$ obtain $R(c)$ and are indifferent between the pool and a refused deviation; higher-cost types settle and strictly prefer the pool. Apply Lemma 3 with $s = 0$, $\lambda_s = 1$, and $w = 0$. Weak-type beliefs support resistance at unused $d \leq h$, while belief on $c = 0$, whose payoff equals $R(0)$, supports resistance at $d > h$. This equilibrium satisfies the criterion and has $T = \Omega = 0$, $q = F(p)$, and $K = I(p)$.

Lemma 2 establishes $x_E < x_P$ and (4) at the maximal-demand boundary. Implicitly differentiating (10) with respect to γ gives (14). Its numerator is $x_E[t_E + t_E + c_2] > 0$, and D_E is positive at the positive root, so the derivative is negative. Equation (15) follows immediately. $\square$

H. EQUILIBRIA IN WHICH EVERY DEMAND IS RESISTED

Both games admit equilibria in which every demand is resisted. Map types injectively into demands in $(h, 1]$ and let the target resist after inferring the type. If compliance would lead to attack, resistance avoids the additional loss γx . If compliance would lead to settlement, a

demand above h costs more than ordinary war. Types above r back down after resistance. Every type receives its resistance continuation value, so none gains by mimicking another resisted signal. Apply [Lemma 3](#) with $\lambda_s = 0$ and $w = -a$. After every unused demand, including zero, assign belief to a weak type and resist. At $d \leq h$, weak types could strictly gain from acceptance and survive the criterion. At $d > h$, every type survives because refusal is the only rational response and gives its equilibrium payoff. Thus these equilibria also satisfy the Intuitive Criterion in both games.

I. POOLING OUTCOME COMPARISONS

Proof. Equations (8) and (11) give

$$\Omega_E = x_E(1 - F(t_E)) < x_E < x_P = \Omega_P$$

At a common $\lambda > 0$, $x_E < x_P$ and $t_E > t_P$ give

$$T_E(\lambda) < \lambda x_P(1 - F(t_P)) = T_P(\lambda)$$

Subtracting the corresponding formulas in the two games gives

$$q_E(\lambda) - q_P(\lambda) = \lambda[F(t_E) - F(t_P)] > 0$$

$$K_E(\lambda) - K_P(\lambda) = \lambda[I(t_E) - I(t_P)] > 0$$

It remains to compare the coercer's expected payoff. At the protected boundary, the target resists against types below t_P and is indifferent between compliance and resistance against the upper tail. At the exposed boundary, it is indifferent between compliance and resistance under the prior. Its ex ante expected payoff is therefore the same in both regimes: the payoff from resisting against the prior.

In either game, total surplus at the maximal pooling demand is

$$V_j(\lambda) = 1 - K_j(\lambda) - (1 - \lambda)a[1 - F(r)], \quad j \in \{P, E\}$$

The last term is the audience cost paid when resistance is followed by backing down. It is identical across games at a common compliance rate. Since the target also receives the same expected payoff,

$$V_E(\lambda) - V_P(\lambda) = U_E(\lambda) - U_P(\lambda) = -\lambda[I(t_E) - I(t_P)] < 0$$

Direct integration gives (17), (18), and the protected mixed-response payoff above. Thus the welfare and payoff comparisons hold for every common $\lambda > 0$.

Implicit differentiation gives (16). Write $s = 1 - \gamma$, $y = sx_E$, and $H = 2p + c_2$. The exposed-demand equation becomes

$$y^2 + \left[\frac{\bar{c}}{s} - H \right] y - ha = 0$$

Thus y is the protected-demand root evaluated at the larger effective support $\frac{\bar{c}}{s}$, so $0 < y < x_P$. Direct simplification gives

$$\frac{\partial x_P}{\partial p} - \frac{\partial x_E}{\partial p} = \frac{2(\bar{c} - H - a)(x_P - y) + \gamma(2x_P + a)(H - 2y)}{D_P D_E}$$

Below $\bar{p}$, both terms in the numerator are positive because $y < x_P < p$. Hence $\frac{\partial x_E}{\partial p} < \frac{\partial x_P}{\partial p}$.

The power derivative of y is $\frac{2y+a}{[2y+\frac{\bar{c}}{s}-H]}$. A second direct comparison gives

$$\frac{\partial x_P}{\partial p} - \frac{\partial y}{\partial p} = \frac{2\left(\frac{\bar{c}}{s} - H - a\right)(x_P - y) + (\bar{c}\frac{\gamma}{s})(a + 2y)}{D_P [2y + \frac{\bar{c}}{s} - H]} > 0$$

Since $t_E = p - y$ and $t_P = p - x_P$, this establishes

$$\frac{\partial F(t_E)}{\partial p} > \frac{\partial F(t_P)}{\partial p} > 0$$

below $\bar{p}$.

Therefore

$$\frac{\partial \Omega_E}{\partial p} = \left(\frac{\partial x_E}{\partial p} \right) (1 - F(t_E)) - x_E \left(\frac{\partial F(t_E)}{\partial p} \right) < \frac{\partial x_P}{\partial p} = \frac{\partial \Omega_P}{\partial p}$$

The benchmark derivative is positive. The exposed derivative is also positive. Substituting the exposed-demand equation into its numerator gives

$$\bar{c}(1 - \gamma) \frac{\partial \Omega_E}{\partial p} = \frac{a(\bar{c} - H) + 2y(\bar{c} - p + a) + 3y^2}{2y + \frac{\bar{c}}{1-\gamma} - H} > 0$$

As λ ranges over $(0, 1]$, (12) spans the interval with lower bound $F(t_E)$ and upper limit $F(r)$. Below $\bar{p}$, both bounds rise more rapidly than $F(t_P)$: the lower-bound result follows from the cutoff-derivative comparison above, while $\frac{\partial F(r)}{\partial p} = \frac{1}{\bar{c}} > \frac{\partial F(t_P)}{\partial p}$ gives the upper-bound result.

Likewise, (13) spans the interval with lower bound $I(t_E)$ and upper limit $I(r)$. Below $\bar{p}$, the lower bound rises more rapidly than $I(t_P)$ because both its level and its cutoff derivative are larger. The upper bound does so because $r > t_P$ and $\frac{\partial t_P}{\partial p} < 1$. Finally,

$$\frac{\partial t_E}{\partial p} = \frac{\bar{c} - (1 - \gamma)(2p + c_2 + a)}{D_E}$$

so the exposed lower bounds turn at $\bar{p}_E = \frac{[\frac{\bar{c}}{1-\gamma} - c_2 - a]}{2}$. These are derivatives of the endpoints of the equilibrium ranges. A realized equilibrium with λ_j varying with p also has the response term $-(F(r) - F(t_j)) \frac{d\lambda_j}{dp}$ in its war-risk derivative, and the analogous conflict-cost term. The Intuitive Criterion leaves those response changes undetermined because the entire pooling family satisfies Lemma 3. This proves the pooling parts of Proposition 3 through Proposition 5 without imposing a direction on equilibrium selection as power changes. $\square$

J. INDEPENDENT COMPLIANCE RATES AT MAXIMAL POOLING DEMANDS

Matching compliance probabilities is consequential because the Intuitive Criterion permits their independent selection. For example, let $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\bar{c} = 2$, and $\gamma = 0.10$, which satisfy Assumptions 1–2. The maximal demands are $x_P \approx 0.19146$ and $x_E \approx 0.17782$. Choosing $\lambda_P = 0.10$ and $\lambda_E = 1$ gives

$$q_P(\lambda_P) \approx 0.42543 > 0.21998 \approx q_E(\lambda_E), \quad T_P(\lambda_P) \approx 0.01524 < 0.13870 \approx T_E(\lambda_E)$$

$$V_P(\lambda_P) \approx 0.65657 < 0.94721 \approx V_E(\lambda_E)$$

Both equilibria satisfy the Intuitive Criterion by Lemma 3. The common-response comparisons therefore identify assurance effects conditional on the matched response, not an ordering of all equilibria satisfying the criterion.

K. TWO-DEMAND SORTING EQUILIBRIA

This section gives the complete characterization with pure demand strategies and two strictly positive demands, each chosen by a positive mass of types and receiving positive compliance. Let $0 < x < y \leq 1$, and let $\alpha > 0$ and $\beta > 0$ denote the target's compliance probabilities after y and x . Necessarily, $\alpha < \beta$. To see why, first note that any type assigned to x must obtain a nonnegative compliance gain: it can otherwise demand more than h , induce resistance regardless of the target's belief, and obtain its resistance continuation payoff. Moreover, the compliance gain is strictly increasing in the demand. If $\alpha \geq \beta$, therefore, every type assigned to x would strictly prefer to mimic y . A two-demand equilibrium thus requires $0 < \alpha < \beta \leq 1$. Write $\rho = \frac{\alpha}{\beta} < 1$. For game $j \in \{P, E\}$, define the sender's gain from compliance rather than resistance as

$$G_P(d, c) = \begin{cases} d - p + c & \text{if } c < r \\ d + a & \text{if } c \geq r \end{cases}$$

$$G_E(d, c) = \begin{cases} \max\{\gamma d, d - p + c\} & \text{if } c < r \\ d + a & \text{if } c \geq r. \end{cases}$$

A type chooses y rather than x exactly when

$$\rho G_j(y, c) \geq G_j(x, c) \tag{19}$$

Under protected compliance, compare the gains directly:

$$\rho G_P(y, c) - G_P(x, c) = \begin{cases} \rho y - x + (1 - \rho)(p - c) & \text{if } c < r \\ \rho(y + a) - (x + a) & \text{if } c \geq r. \end{cases}$$

This difference is strictly decreasing below r and constant above it. Its nonnegative set therefore has at most one cutoff. The argument also covers $c_2 = 0$ without dividing by a zero compliance gain. Under exposed compliance, both gains are strictly positive. Define $t_d = p - (1 - \gamma)d$. Then

$$\frac{G_E(x, c)}{G_E(y, c)} = \begin{cases} \frac{x}{y} & \text{if } c \leq t_y \\ \frac{\gamma x}{y - p + c} & \text{if } t_y < c < t_x \\ \frac{x - p + c}{y - p + c} & \text{if } t_x \leq c < r \\ \frac{x + a}{y + a} & \text{if } c \geq r. \end{cases}$$

The ratio is first flat, then decreasing, then increasing, and finally flat, on the branches that intersect the type support. This nonmonotonicity permits either one cutoff or two.

Receiver optimality rules out the upper indifference plateau when $x > 0$. If $\rho \geq \frac{x+a}{y+a}$, every resolved type under protection strictly prefers y . Under exposure, every resolved type that would settle after complying with x strictly prefers y . Thus a low-demand pool could contain only weak types, and under exposure also attackers. Compliance costs the target $x > 0$ against a weak type and an additional $\gamma x > 0$ against an attacker. The target would strictly resist. Consequently, every equilibrium in this class has $\rho < \frac{x+a}{y+a}$, and its upper cutoff lies below r .

L. MONOTONE TWO-POOL SORTING

We first characterize bargaining in which larger demands identify coercers with lower war costs. In a monotone profile with $0 < x < y \leq 1$, types $c < k$ demand y and types $c \geq k$ demand x . The target complies with probabilities $\alpha < \beta$. The conditions below establish when this ordering is sustainable and whether acceptance assures peace.

PROPOSITION 7 (When larger demands distinguish more resolved coercers): Suppose there are two positive demands, each chosen by a positive share of types and sometimes accepted.

- In the *protected compliance game*, monotone sorting exists exactly under conditions (i)–(iii) below. All high-demand types settle after acceptance.
- In the *exposed compliance game*, monotone sorting exists exactly under conditions (iv)–(viii) below. Some high-demand types may attack after acceptance.

In both games, the target accepts the high demand less often, and every low-demand type settles after acceptance. If the target strictly prefers accepting the low demand, it accepts for certain. If accepting and resisting give it the same payoff, it can accept with any positive probability, with the high-demand response adjusted in the same proportion.

Exact conditions.

A monotone two-pool equilibrium in the protected compliance game exists exactly under conditions (i)–(iii):

- (i) $y = h$ and $0 < k < r$;
- (ii) $\rho = \frac{x-p+k}{h-p+k} = \frac{x-p+k}{k+c_2}$ lies in $(0, 1)$; and
- (iii) $x(\bar{c} - k) \leq h(r - k)$.

A monotone two-pool equilibrium in the exposed compliance game exists exactly under conditions (iv)–(viii). Let $m_y = \max\{0, t_y\}$. Then

- (iv) $m_y < k < r$ and $t_x < k$;
- (v) $(h - y)(k - m_y) = \gamma y m_y$;
- (vi) $\rho = \frac{x-p+k}{y-p+k}$ lies in $(0, 1)$;
- (vii) $\rho G_E(y, 0) \geq G_E(x, 0)$; and
- (viii) $x(\bar{c} - k) \leq h(r - k)$.

In either game, a strict final inequality implies $\beta = 1$ and $\alpha = \rho$. At equality, any $\beta \in (0, 1]$ and $\alpha = \rho\beta$ is admissible.

The exposed characterization includes the behaviorally assured equilibrium as a boundary case. If no type attacks after receiving the high demand, then $m_y = 0$, condition (v) forces $y = h$, and the additional exposed incentive restrictions reduce to

$$\gamma \leq \frac{c_2}{h}, \quad \rho c_2 \geq \gamma x \tag{20}$$

These are exactly the protected profiles that survive exposure without changing their on-path behavior. When $m_y > 0$, by contrast, low-cost types demand $y < h$ and attack after compliance. The target can still mix because settlements by the remaining high-demand types offset those attacks.

M. PROOF OF THE MONOTONE CHARACTERIZATION

M.1. Proof of Proposition 7

Proof. We prove necessity using sender indifference and target optimality, then apply Lemma 3 for sufficiency. Every equilibrium satisfying the criterion is a PBE, so the usual incentive and posterior restrictions remain necessary. Under protected compliance, the gain difference in (19)

is strictly decreasing below r . The upper-plateau exclusion above gives $k < r$, and both pools have positive mass, so $k > 0$. Cutoff indifference gives

$$\rho = \frac{x - p + k}{y - p + k}$$

Every type in the high pool fights after resistance and settles after compliance. Since $0 < \alpha < \beta \leq 1$, the target mixes after y , which requires $y = h$. Its cutoff gain $h - p + k = k + c_2$ is strictly positive, so the displayed ratio is well defined. At the low signal, the target's compliance surplus, multiplied by the posterior denominator, is $h(r - k) - x(\bar{c} - k)$, which yields the final inequality. The decreasing gain difference assigns exactly the types below k to y . Types above r strictly prefer x because cutoff indifference implies $(x + a) > \rho(h + a)$. Every on-path gain is nonnegative: $h - p + c = c_2 + c \geq 0$ in the high pool, and $x - p + k > 0$ at the bottom of the low pool. This verifies necessity and sender optimality for sufficiency.

Under exposed compliance, the high-demand pool contains attackers $[0, m_y)$ and settlers $[m_y, k)$. The target's unnormalized payoff difference between compliance and resistance after y is

$$(h - y)(k - m_y) - \gamma y m_y$$

Because $0 < \alpha < 1$, this expression must equal zero. A high pool consisting only of attackers would make it negative, so $m_y < k$. A nondegenerate upper cutoff must lie on the increasing branch of the ratio, giving $t_x < k$ and condition (vi). The ratio is weakly decreasing before t_x and increasing afterward up to r . Thus the bottom type supplies the only additional incentive constraint, condition (vii), while the cutoff supplies the upper boundary. The low signal contains only settlers after compliance, so target optimality again gives $x(\bar{c} - k) \leq h(r - k)$.

Conversely, conditions (iv)–(viii) make the target indifferent after y and weakly willing to comply after x . The shape of the ratio and the bottom and cutoff inequalities assign every $c < k$ to y and every $c \geq k$ to x . Types $c < m_y$ attack after high-demand compliance, those in $[m_y, k)$ settle, and every type assigned to x settles because $t_x < k$. Exposed compliance gains are strictly positive, and the protected gains were shown to be nonnegative. Both profiles therefore meet [Lemma 3](#), with $s = x$, $\lambda_s = \beta$, and $w = \beta x - (1 - \beta)a$. Its off-path prescription

completes both games for every admissible β , including receiver-indifference boundaries. Under protection with $c_2 = 0$, the zero-cost type has $v_P(0) = R(0)$; belief on that type and compliance with an unused $d < w$ satisfy the criterion without creating a profitable deviation. This proves necessity and sufficiency under the stated solution concept. $\square$

N. LOW DEMANDS CAN FEIGN RESTRAINT ONLY UNDER EXPOSED COMPLIANCE

We next characterize bargaining in which the same low demand attracts coercers planning an attack and coercers seeking to avoid resistance. In a two-sided profile with strictly positive demands, an interior interval $[b, k)$ demands y , while the belligerent and weak tails both demand x . This pattern is impossible under protected compliance because the high-demand gain difference decreases with the war cost. It becomes possible only under exposed compliance, which makes acceptance valuable to the most belligerent types as well as to the weakest types. The characterization covers both strict preferences between demands and cases in which groups of coercers are indifferent.

PROPOSITION 8 (When low demands combine belligerent and weak coercers): Suppose there are two positive demands, each chosen by a positive share of types and sometimes accepted.

- In the *protected compliance game*, two-sided sorting is impossible.
- In the *exposed compliance game*, the necessary and sufficient conditions below distinguish two forms. In regular sorting, low-cost attackers and high-cost types choose the low demand, while an intermediate group demands more and settles after acceptance. In the other form, some low-cost types are indifferent between demands. Any subset of that indifferent group with the required probability can form the bottom part of the low-demand pool; the interval assignment below is one representation.

Exact conditions.

With two strictly positive demands, protected compliance has no two-sided two-pool PBE. Exposed compliance has the following two classes.

First, a regular two-sided equilibrium exists if and only if $y = h$, $0 < x < p$, and

$$\rho_{\min} < \rho < \min\{\rho_0, \rho_r\}$$

where

$$t_h = \gamma h - c_2, \quad t_x = p - (1 - \gamma)x$$

$$\rho_{\min} = \frac{\gamma x}{t_x + c_2}, \quad \rho_0 = \frac{\gamma x}{\max\{\gamma h, c_2\}}, \quad \rho_r = \frac{x + a}{h + a}$$

The cutoffs are

$$b = \frac{\gamma x}{\rho} - c_2, \quad k = \frac{p - x + \rho c_2}{1 - \rho}$$

they satisfy $\max\{0, t_h\} < b < t_x < k < r$, and target optimality requires

$$x(\bar{c} - k + \gamma b) \leq h(r - k) \tag{21}$$

Second, a tie-supported two-sided equilibrium exists if and only if $0 < x < y < h$, $t_y > 0$, $\rho = \frac{x}{y}$, $k = p$, and

$$b = t_y - \frac{(h - y)(p - t_y)}{\gamma y}$$

lies in $(0, t_y)$, while

$$x(\bar{c} - p) + \gamma x b \leq h a$$

In a canonical representation, types in $[0, b)$ and $[p, \bar{c}]$ demand x , while types in $[b, p)$ demand y . More generally, any measurable subset of $[0, t_y]$ with total length b , hence probability $\frac{b}{\bar{c}}$, can form the lower low-demand pool. Boundary equalities give the corresponding weak and degenerate profiles.

In the regular class, low-cost types below b attack after low-demand compliance, types in $[b, k)$ settle after high-demand compliance, and types above k settle after the low demand. If the target's final inequality is strict, $\beta = 1$ and $\alpha = \rho$; at equality, any $\beta \in (0, 1]$ with $\alpha = \rho\beta$

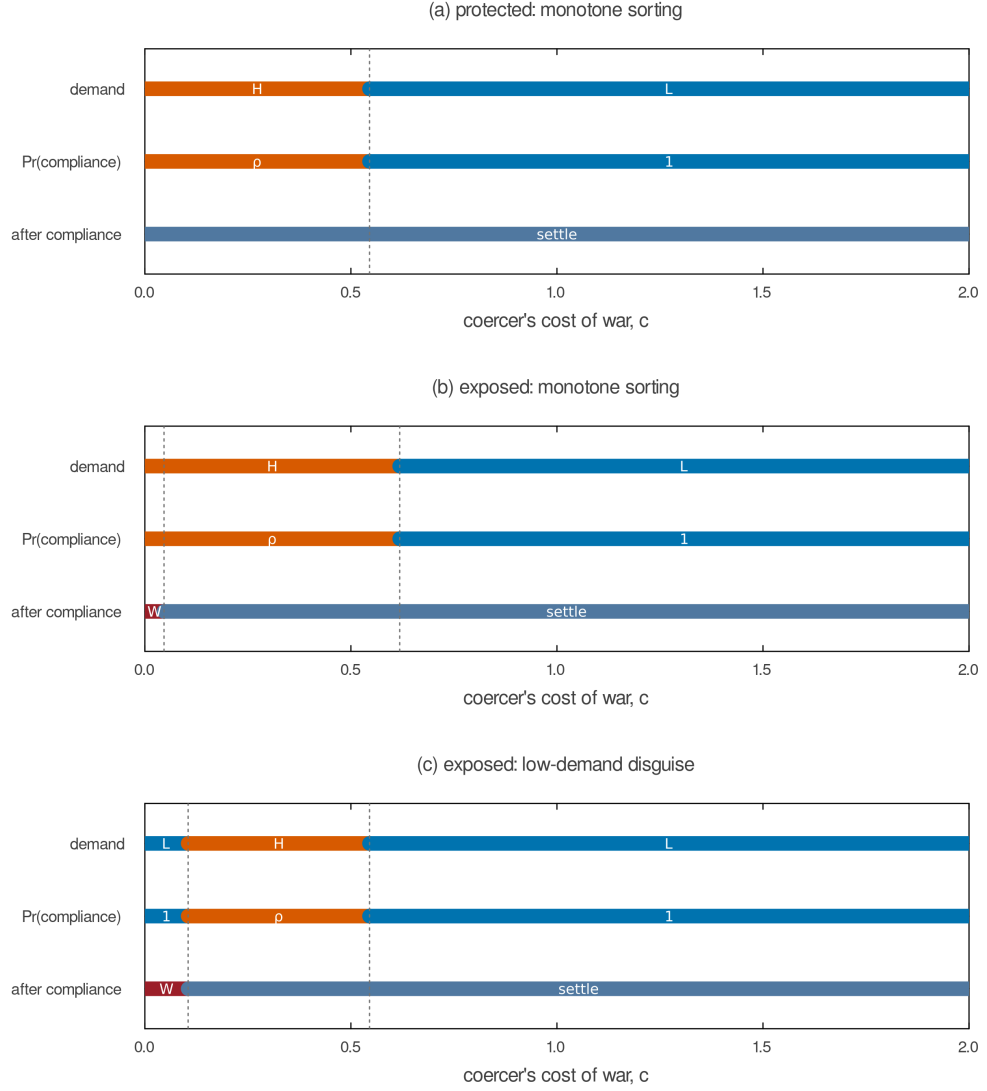


Figure 6: Only exposed compliance lets belligerent types feign restraint with a low demand

Note: Panel (a) shows the monotone ordering required under protected compliance. Panel (b) shows monotone sorting under exposed compliance when the lowest-cost members of the high-demand pool attack after compliance. Panel (c) shows the additional ordering available only under exposed compliance: belligerent and weak types pool at the low demand, leaving moderate types at the high demand. Panels (a) and (c) use $y = h = 0.62$ and $\rho = 0.08$; panel (b) uses $y = 0.615$ and $\rho = 0.187$. Other parameters are $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.10$, $\bar{c} = 2.00$, and $x = 0.10$.

works. In the tie-supported class, all types below t_y are indifferent between the two signals.

The specified assignment divides this plateau so that the target is willing to mix after y .

O. PROOF OF THE FEIGNED-RESTRAINT CHARACTERIZATION

O.1. Proof of Proposition 8

Proof. Under protected compliance, the gain difference $\rho G_P(y, c) - G_P(x, c)$ is strictly decreasing for $c < r$ and constant above r . The high-demand set is its nonnegative set, which cannot be an interior interval. Two-sided sorting is impossible.

Under exposed compliance, the ratio displayed before Proposition 7 is flat, decreasing, increasing, and flat over its four branches. For a regular two-sided profile, ρ must cross it once on the decreasing branch and once on the increasing branch. The endpoint restrictions are exactly

$$\rho_{\min} < \rho < \min\{\rho_0, \rho_r\}$$

Solving the two equalities yields b and k and implies $\max\{0, t_h\} < b < t_x < k < r$. Because every high-pool type fights after resistance and settles after compliance, target mixing forces $y = h$. At the low signal, attackers $[0, b)$ impose the extra exposure loss $\gamma x b$, settlers $[k, r)$ avert war, and types above r accept a transfer they could otherwise avoid. The target's unnormalized compliance surplus is therefore

$$h(r - k) - x(\bar{c} - k + \gamma b)$$

which proves the regular condition.

The upper flat branch cannot support indifference between two positive-compliance demands, as established before Proposition 7. A positive-mass indifference set can therefore arise only on the bottom flat branch, where the ratio equals $\frac{x}{y}$. Hence $\rho = \frac{x}{y}$ and $t_y > 0$. Above t_x , cutoff indifference gives $k = p$. Let types $[0, b)$ use x and $[b, t_y]$ use y . Target indifference after y requires

$$(h - y)(p - t_y) = \gamma y(t_y - b)$$

which yields the stated b . The target's low-signal compliance condition is $x(\bar{c} - p) + \gamma x b \leq h a$. These restrictions are also sufficient because every type below t_y is indifferent, types in (t_y, p) strictly prefer y , and types above p prefer x .

In both exposed classes, on-path compliance gains are strictly positive. The on-path incentive and target-optimality conditions therefore meet [Lemma 3](#), with $s = x$, $\lambda_s = \beta$, and $w = \beta x - (1 - \beta)a$. Its off-path prescription completes both classes under the Intuitive Criterion, including mixed responses and arbitrary assignments of indifferent bottom types. The protected impossibility follows already from PBE incentives. The strict classes, their boundary equalities, and the positive-mass bottom plateau exhaust the ways a flat-decreasing-increasing-flat ratio can generate a two-sided assignment. $\square$

P. AN INTERIOR FEIGNED-RESTRAINT EXAMPLE

The regular class includes equilibria with demands and acceptance rates bounded away from zero. Set $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.30$, $\bar{c} = 1.50$, $x = 0.20$, and $\rho = 0.20$, with $y = h = 0.62$. All maintained assumptions hold, and the regular conditions are strict:

$$\rho_{\min} = \frac{1}{8} < \rho = \frac{1}{5} < \rho_0 = \frac{10}{31} < \rho_r = \frac{25}{46}$$

$$0 < t_h = 0.166 < b = 0.28 < t_x = 0.46 < k = 0.505 < r = 0.90$$

The target's unnormalized surplus from accepting the low demand is

$$h(r - k) - x(\bar{c} - k + \gamma b) = 0.0291 > 0$$

so $\beta = 1$ and $\alpha = \rho$. The belligerent low-demand group, high-demand group, and upper low-demand group have masses $\frac{14}{75}$, $\frac{3}{20}$, and $\frac{199}{300}$, respectively. The upper low-demand group includes weak types with $c \geq r$, whose population mass is $\frac{2}{5}$.

For these parameters, the difference between the coercer's expected payoffs from the high and low demands is

$$D_P(c) = \begin{cases} 0.404 - 0.8c & \text{if } c < 0.90 \\ -0.316 & \text{if } c \geq 0.90 \end{cases}$$

under protection, and

$$D_E(c) = \begin{cases} -0.0228 & \text{if } c \leq 0.166 \\ -0.056 + 0.2c & \text{if } 0.166 < c < 0.46 \\ 0.404 - 0.8c & \text{if } 0.46 \leq c < 0.90 \\ -0.316 & \text{if } c \geq 0.90 \end{cases}$$

under exposure. The protected profile assigns types below k to the high demand; the exposed profile assigns only types between b and k to it. The target is indifferent after the high demand in both games. Its unnormalized low-demand compliance surplus under protection is $0.0459 > 0$, so the same demands and acceptance rates support the protected counterpart. Every type strictly exceeds its resistance payoff, and [Lemma 3](#) completes both profiles with resistance after unused demands.

The example extends to an open region. Allow independent perturbations of at most 0.001 in each of $p, c_2, a, \gamma, \bar{c}, x, \rho$, while setting $y = p + c_2$, $\alpha = \rho$, and $\beta = 1$. Rational interval bounds preserve all maintained assumptions and regular-class inequalities throughout this box. In particular, $x, \rho \geq 0.199$, the exposed high-demand group has mass greater than 0.14428, and the exposed low-demand compliance surplus exceeds 0.02264. The interior of the box therefore supplies an open set with substantial demands, acceptance rates, and group masses. This is a local existence result, with no claim about their magnitudes throughout the parameter space.

At the center of the box, the matched comparison gives

$$\begin{aligned}\Delta P_C &= \frac{56}{375}, & \Delta T &= -\frac{217}{9375}, & \Delta q &= \frac{14}{375} \\ \Delta U &= \frac{49}{9375}, & \Delta U_2 &= -\frac{7}{625}, & \Delta V &= -\frac{56}{9375}\end{aligned}$$

Thus exposure increases compliance by about 15 percentage points while reducing the peaceful transfer. Belligerent types gain from the added attack option, and the target loses both this gain and the additional joint conflict costs.

Q. ZERO-DEMAND SORTING BOUNDARIES

The restriction to strictly positive demands matters because the target can accept a zero demand from weak types. For example, let $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\bar{c} = 2$, and $\gamma = 0.10$, with $x = 0$, $y = 0.61$, $\beta = 1$, and $\alpha = \rho = \frac{30}{91} = \frac{a}{y+a}$. In the protected game, set $k_P = \frac{279}{305} \approx 0.91475 > r = 0.90$. In the exposed game, set $k_E = \frac{554379}{610000} \approx 0.90882 > r$, and note that $t_y = 0.051$. In each game, assign $c < k_j$ to y and $c \geq k_j$ to zero. All resolved types strictly prefer y ;

every weak type is indifferent between the two demands, receiving zero from either. The target is indifferent after the high demand because

$$yk_P = hr, \quad yk_E + (t_y + c_2)t_y = hr$$

It is also indifferent after the zero demand, whose pool contains only weak types. Every type receives at least $R(c)$, so Lemma 3 applies with $s = 0$, $\lambda_s = 1$, and $w = 0$. Weak-type beliefs support resistance at unused $d \leq h$. At $d > h$, every type receives strictly more than $R(c)$ on path, so the criterion is vacuous and weak-type beliefs again support resistance. These equilibria satisfy the Intuitive Criterion and have an upper indifference plateau and $k_j > r$; they lie outside the positive-demand characterizations above.

R. PROOF OF EQUILIBRIUM-CLASS EXISTENCE

R.1. Proof of Proposition 1

Proof. The proof proceeds by construction and then establishes necessity for reassuring separation. Lemma 2 and the protected-compliance pooling construction give a positive-compliance pool at x_P for every admissible parameter vector. Lemma 2 and the exposed-compliance pooling construction give an all-type pool at every $x \in (0, x_E]$. Thus pooling exists throughout the maintained parameter space in both games.

For protected monotone sorting, set $y = h$, $k = p$, and $\rho = \frac{x}{h}$. Choose

$$0 < x < \min\left\{h, \frac{ha}{\bar{c} - p}\right\}$$

Then $0 < k < r$, $\rho \in (0, 1)$, and

$$\rho = \frac{x - p + k}{h - p + k} = \frac{x}{h}, \quad x(\bar{c} - k) < h(r - k) = ha$$

These are the protected conditions in Proposition 7, so a reassuring monotone two-pool equilibrium exists. The protected gain difference is decreasing below r and constant above it. Its high-demand set can therefore have only one cutoff, which rules out feigned restraint.

Consider next a reassuring monotone equilibrium under exposed compliance. Because no high-demand type attacks after compliance, $m_y = 0$. Target indifference after the high demand and [Proposition 7](#) imply $(h - y)k = 0$. Since $k > 0$, $y = h$. No attack after high-demand compliance then requires

$$t_h = p - (1 - \gamma)h = \gamma h - c_2 \leq 0$$

or equivalently $\gamma \leq \frac{c_2}{h}$. This proves necessity. For sufficiency, suppose $\gamma \leq \frac{c_2}{h}$ and again set $y = h$, $k = p$, and $\rho = \frac{x}{h}$, with x satisfying the strict bound above. Then $m_y = 0$, $t_x < k$, and

$$\rho G_E(h, 0) = \frac{xc_2}{h} \geq \gamma x = G_E(x, 0)$$

The remaining exposed conditions in [Proposition 7](#) coincide with the protected conditions just verified. Hence reassuring monotone sorting exists if and only if $\gamma \leq \frac{c_2}{h}$.

It remains to construct a feigned-restraint equilibrium for every admissible $\gamma > 0$. Let

$$L = \max\{0, \gamma h - c_2\}$$

Because $\gamma < 1$, $L < p$. Choose $b \in (L, p)$. For $x > 0$, define

$$\rho = \frac{\gamma x}{b + c_2}, \quad t_x = p - (1 - \gamma)x, \quad k = \frac{p - x + \rho c_2}{1 - \rho}$$

This choice makes $b = \gamma \frac{x}{\rho} - c_2$. Since $b > L$, it also gives

$$b + c_2 > \max\{\gamma h, c_2\}$$

and therefore $\rho < \rho_0$. As x approaches zero, ρ approaches zero, t_x and k approach p , ρ_r approaches $\frac{a}{h+a} > 0$, and

$$h(r - k) - x(\bar{c} - k + \gamma b)$$

approaches $ha > 0$. Since $b < p < r$, continuity permits a sufficiently small $x > 0$ such that $b < t_x$, $\rho < \min\{1, \rho_r\}$, $k < r$, and the low-demand compliance inequality is strict. Moreover,

$$\rho = \frac{\gamma x}{b + c_2} > \frac{\gamma x}{t_x + c_2} = \rho_{\min}$$

and this last inequality is equivalent to $t_x < k$. All strict conditions for the regular feigned-restraint equilibrium in [Proposition 8](#) therefore hold. Such an equilibrium exists at every admissible parameter vector with $\gamma > 0$.

Finally, the demand-choice condition assigns a type to the high demand when $\rho G_j(y, c) \geq G_j(x, c)$. Under protection, the gain difference has one crossing. Under exposure, the gain ratio is flat, decreasing, increasing, and flat; its sublevel set at ρ is either a lower interval or an interior interval. Receiver optimality excludes indifference on the upper plateau, while bottom-plateau ties are covered by [Proposition 8](#). Only indifferent types can be assigned in other ways. The two orderings therefore exhaust the class with pure demand strategies, two strictly positive demands, positive mass at both signals, and positive compliance at each. Every existence construction above satisfies the Intuitive Criterion by [Lemma 3](#), while each impossibility follows from necessary PBE conditions. Thus the equilibrium-class characterization holds under the stated solution concept. $\square$

S. TWO-POOL OUTCOME FORMULAS

Let P_C denote the ex ante probability of compliance. In a monotone equilibrium, let m_y be the length of the interval of high-demand types that attack after compliance; its prior probability is $\frac{m_y}{\bar{c}}$. Then

$$P_C^M = \frac{\alpha k + \beta(\bar{c} - k)}{\bar{c}}$$

$$T_M = \frac{\alpha y(k - m_y) + \beta x(\bar{c} - k)}{\bar{c}}, \quad \Omega_M = \frac{T_M}{P_C^M}$$

$$q_M = \frac{m_y + (1 - \alpha)(k - m_y) + (1 - \beta)(r - k)}{\bar{c}}$$

$$K_M = I(m_y) + (1 - \alpha)(I(k) - I(m_y)) + (1 - \beta)(I(r) - I(k))$$

For protected compliance, set $m_y = 0$ and $y = h$. For the behaviorally assured exposed subclass, set $m_y = 0$ as well.

For a regular two-sided equilibrium,

$$P_C^T = \frac{\alpha(k - b) + \beta(b + \bar{c} - k)}{\bar{c}}$$

$$T_T = \frac{\alpha h(k - b) + \beta x(\bar{c} - k)}{\bar{c}}, \quad \Omega_T = \frac{T_T}{P_C^T}$$

$$q_T = \frac{b + (1 - \alpha)(k - b) + (1 - \beta)(r - k)}{\bar{c}}$$

$$K_T = I(b) + (1 - \alpha)(I(k) - I(b)) + (1 - \beta)(I(r) - I(k))$$

For the tie-supported class, replace k by p in P_C^T , use

$$T_T = \frac{\alpha y(p - t_y) + \beta x(\bar{c} - p)}{\bar{c}}$$

$$q_T = \frac{t_y + (1 - \alpha)(p - t_y) + (1 - \beta)(r - p)}{\bar{c}}$$

$$K_T = I(t_y) + (1 - \alpha)(I(p) - I(t_y)) + (1 - \beta)(I(r) - I(p))$$

T. PROOFS OF THE MAIN OUTCOME PROPOSITIONS

T.1. Proof of [Proposition 3](#) through [Proposition 5](#)

Proof. The pooling comparison above establishes the first part of each proposition. For separation, [Proposition 7](#) establishes that protected monotone sorting requires $y = h$. Under exposed monotone sorting, target indifference requires $(h - y)(k - m_y) = \gamma y m_y$, with $k > m_y$. Hence $y = h$ when $m_y = 0$ and $y < h$ exactly when $m_y > 0$.

[Proposition 8](#) proves that protected compliance has no two-sided two-pool PBE and gives necessary and sufficient conditions for the exposed regular class. This class contains an open set. For example, at $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\bar{c} = 2$, $\gamma = 0.10$, $x = 0.10$, $y = h = 0.62$, and $\rho = 0.08$, all inequalities in [Proposition 8](#) are strict: $0.042 < b = 0.105 < t_x = 0.51 < k = 0.545 < r = 0.90$, and the low-demand compliance inequality has left side 0.147 and right side 0.220. Continuity preserves the inequalities in a neighborhood of these parameters.

For every regular equilibrium in the exposed compliance game, the two sender-indifference conditions imply

$$\rho = \frac{\gamma x}{c_2 + b} \quad \wedge \quad \rho = \frac{x - p + k}{c_2 + k}$$

Solving gives

$$b = \frac{\gamma x}{\rho} - c_2, \quad k = \frac{p - x + \rho c_2}{1 - \rho}$$

Holding x and ρ fixed while remaining in the regular branch gives $d_{\frac{b}{d}}\gamma = \frac{x}{\rho} > 0$ and $d_{\frac{k}{d}}\gamma = 0$.

Let A denote post-compliance attack. Then

$$\Pr(A \text{ mid } x, C) = \frac{b}{b + \bar{c} - k}, \quad \frac{d \Pr(A \text{ mid } x, C)}{d\gamma} = \text{frac}\{x(\bar{c} - k), \rho(b + \bar{c} - k)^2\} > 0$$

The left side of the target's low-demand compliance condition (21) has derivative

$$\frac{d}{d\gamma} x(\bar{c} - k + \gamma b) = x \left(b + \frac{\gamma x}{\rho} \right) > 0$$

while its right side is constant. Thus greater exposure makes the low demand less reassuring and its compliance condition harder to satisfy.

For the outcome comparison, take a regular two-sided equilibrium in the exposed compliance game in which the low-demand inequality is strict, so $\beta = 1$ and $\alpha = \rho$. Its protected counterpart uses the same h, x, ρ , and upper cutoff k . Under protection, the high-demand settlement set is $\mathcal{H}_P = [0, k)$. Under exposed compliance it is $\mathcal{H}_E = [b, k)$. Because types are uniform on $[0, \bar{c}]$, their probability masses are $\frac{k}{\bar{c}}$ and $\frac{k-b}{\bar{c}}$. The exposed mass therefore falls at rate

$$\frac{d}{d\gamma} \frac{k - b}{\bar{c}} = -\frac{x}{\rho \bar{c}} < 0$$

The types removed from this set enter the low-demand pool and attack after compliance. The target is indifferent after h in both games. At the low demand, removing the exposure loss $\gamma x b$ makes compliance strictly optimal under protection, so the same target responses are sequentially rational.

Let Δ denote the exposed outcome minus the protected outcome. Only types $c \in [0, b)$ change signals. Using the formulas above gives

$$\Delta P_C = \frac{(1-\rho)b}{\bar{c}} > 0, \quad \Delta T = -\frac{\rho hb}{\bar{c}} < 0$$

$$\Delta q = \frac{\rho b}{\bar{c}} > 0, \quad \Delta K = \rho I(b) > 0$$

For coercive ability, define

$$L_P = \bar{c} - (1-\rho)k, \quad N_P = \rho hk + x(\bar{c} - k)$$

and $L_E = L_P + (1-\rho)b$, $N_E = N_P - \rho hb$. Then

$$\Delta \Omega = -\frac{b(\rho h L_P + (1-\rho)N_P)}{L_P L_E} < 0$$

Under protection, type $c < b$ receives $p - c + \rho(c + c_2)$. Under exposure it receives $p - c + \gamma x$.

The lower cutoff condition gives $\gamma x = \rho(b + c_2)$. Integration therefore yields

$$\Delta U = \int_0^b \rho(b - c) dF(c) = \frac{\rho b^2}{2\bar{c}} > 0$$

The target's payoff changes by

$$\Delta U_2 = -\frac{\gamma x b}{\bar{c}} = -\frac{\rho b(b + c_2)}{\bar{c}} < 0$$

Hence the change in total surplus is

$$\Delta(U + U_2) = -\rho I(b) < 0$$

These calculations prove the separating parts of [Proposition 3](#) through [Proposition 5](#). Both members of each matched comparison satisfy the Intuitive Criterion by [Lemma 3](#), whose completion changes no on-path payoff or response. Hence all reported comparisons hold under the stated solution concept. $\square$

U. PURE COMPLIANCE AS A SPECIAL CASE

Set $\lambda_P = \lambda_E = 1$ at the maximal pooling demands. Then $q_j = F(t_j)$ and $K_j = I(t_j)$ for $j \in \{P, E\}$. The protected and exposed cutoff derivatives are

$$\frac{\partial q_P}{\partial p} = \frac{\bar{c} - 2p - c_2 - a}{\bar{c} D_P}, \quad \frac{\partial q_E}{\partial p} = \frac{\bar{c} - (1-\gamma)(2p + c_2 + a)}{\bar{c} D_E}$$

They generate distinct turning points:

$$\bar{p}_P = \frac{\bar{c} - c_2 - a}{2}, \quad \bar{p}_E = \frac{\frac{\bar{c}}{1-\gamma} - c_2 - a}{2} > \bar{p}_P \quad (22)$$

Below $\bar{p}_P$, war rises faster under exposed than protected compliance. Between the two thresholds, protected-compliance war falls while exposed-compliance war continues to rise. Above $\bar{p}_E$, both derivatives are negative. Expected conflict cost inherits these signs because $K_j = I(t_j)$ and $I'(t) > 0$; below $\bar{p}_P$, it also rises faster under exposed compliance because $t_E > t_P$ and $\frac{\partial t_E}{\partial p} > \frac{\partial t_P}{\partial p} > 0$.

V. BARGAINING ADJUSTMENT AND ATTAINABLE OUTCOMES

This section proves [Proposition 6](#) and characterizes the marginal outcome ranges when demands and target responses can adjust in both games. The result maintains the uniform prior and pure demand strategies. The two-demand proof below establishes the bounds and their sharpness; [Lemma 8](#) extends the inequalities to every finite set of strictly positive on-path accepted demands, each chosen by positive prior mass. This includes all three-demand equilibria in [Proposition 2](#). Accepted zero demands are excluded. A marginal range describes one outcome at a time; the ranges do not specify which combinations are jointly attainable.

Write $g_P = 0$, $g_E = \gamma$, and $t_j(x) = p - (1 - g_j)x$. Define

$$\Psi_j(x) = hr - \bar{c}x - t_j(x)[t_j(x) + c_2] = ha - [\bar{c} - (1 - g_j)(2p + c_2)]x - (1 - g_j)^2 x^2$$

Its unique positive zero is the maximal pooling demand x_j . The maintained assumptions imply $0 < x_E < x_P < p$ and $0 < t_P < t_E < r < \bar{c}$. The favorable pooling endpoints are

$$q_j^* = F(t_j), \quad K_j^* = I(t_j), \quad U_j^* = hF(r) - I(t_j), \quad V_j^* = 1 - I(t_j)$$

where $t_j = t_j(x_j)$. Universal resistance yields

$$q_R = F(r), \quad K_R = I(r), \quad U_R = -a + \frac{r^2}{2\bar{c}}, \quad V_R = 1 - I(r) - a[1 - F(r)]$$

Within the class of pure demand strategies whose on-path accepted demands form a finite set of at least two strictly positive amounts, each chosen by positive prior mass, the exact marginal ranges in game j are

$$q \in (q_j^*, q_R), \quad K \in (K_j^*, K_R), \quad U \in (U_R, U_j^*), \quad V \in (V_R, V_j^*)$$

Every value in each interval is already attained by a two-demand equilibrium, and no endpoint is attained within the larger finite-demand class. Adding the characterized pooling and all-resisted equilibria closes the four intervals. Because $t_E > t_P$, each exposed interval is a strict subset of its protected counterpart. This is a statement about attainable outcomes, without a rule selecting an equilibrium.

V.1. Proof of Proposition 6

Proof. First take a two-demand profile with certain low-demand acceptance, $\beta = 1$, and high-demand acceptance $\alpha = \rho \in (0, 1)$. In every characterized class, weak types choose the low demand x and settle, and high-demand types are resolved. Backing down therefore never occurs on path.

Let $v(c)$ be the coercer's equilibrium payoff and $w(c)$ its probability of war. Each demand's continuation payoff is a convex, piecewise affine function of the cost. Sender optimality makes v the maximum of these functions, so v is absolutely continuous and $v'(c) = -w(c)$ almost everywhere. In the bottom indifference plateau, both demands induce war and have the same slope; its arbitrary assignment does not affect this conclusion. Since $v(\bar{c}) = x$, integration under the uniform prior gives

$$\bar{c}q = v(0) - x, \quad U = x + \frac{1}{\bar{c}} \int_0^{\bar{c}} cw(c) dc = x + K - c_2q$$

Under protection, the zero-cost type can guarantee resistance and war by demanding more than h , so $v(0) \geq p$. Under exposure, it can imitate the surely accepted low demand and then attack, so $v(0) \geq p + \gamma x$. Thus, writing $z = \bar{c}q$,

$$z \geq t_j(x)$$

At a fixed war probability, conflict costs are minimized by concentrating war among the lowest-cost types. Specifically,

$$\bar{c}[K - I(z)] = \int_0^{\bar{c}} (c - z)[w(c) - \mathbb{1}[c < z]] \, dc \geq 0$$

Both factors have the same sign below and above z . Equality requires $w(c) = \mathbb{1}[c < z]$ almost everywhere. Every positive high-demand pool contains a positive mass of resolved settlers: the target would not mix after a demand whose entire pool attacked after compliance. These settlers fight with probability $1 - \rho \in (0, 1)$, so equality is impossible and $K > I(z)$.

The target can resist either demand, giving

$$U_2 \geq 1 - hF(r)$$

Every coercer can imitate the low demand and obtain at least x . Together with $U + U_2 = 1 - K$, this implies $x \leq h\frac{r}{c} < p$, and hence $t_j(x) > 0$. Combining the envelope identity, the target's option to resist, and the strict conflict-cost bound yields

$$hr \geq \bar{c}(U + K) = \bar{c}x + 2\bar{c}K - c_2z > \bar{c}x + z^2 + c_2z \geq \bar{c}x + t_j(x)^2 + c_2t_j(x)$$

Therefore $\Psi_j(x) > 0$. Assumption 1 makes Ψ_j strictly decreasing on the nonnegative axis, so $x < x_j$. It follows that

$$q \geq \frac{t_j(x)}{\bar{c}} > q_j^*, \quad K > I(\bar{c}q) > K_j^*, \quad U \leq hF(r) - K < U_j^*, \quad V = 1 - K < V_j^*$$

War occurs only among resolved types, and high-demand settlers avoid it with positive probability. Hence $q < q_R$ and $K < K_R$. Every type obtains at least its resistance payoff, and the positive mass of weak types strictly gains, so $U > U_R$. Total welfare also strictly exceeds V_R .

If the original equilibrium has $\beta < 1$, the target is indifferent after both demands. Raising β to one and retaining $\rho = \frac{\alpha}{\beta}$ preserves its best responses and multiplies all between-demand sender gains by the same positive factor. The resulting profile satisfies [Lemma 3](#). For each outcome $Y \in \{q, K, U, V\}$, the original profile satisfies

$$Y(\beta) = \beta Y(1) + (1 - \beta)Y_R$$

Its outcome lies strictly between the pure-low-response outcome and universal resistance. This proves all strict bounds, including mixed low-demand responses.

To establish sharpness and the entire marginal ranges, construct two-demand equilibria with low-demand indifference approaching the favorable pooling endpoint. Under protection, choose $k > t_P$ approaching t_P , and set

$$x = \frac{h(r-k)}{\bar{c}-k}, \quad \rho = \frac{x-p+k}{k+c_2}, \quad y = h$$

At the limit, $x = x_P$ and $\rho = 0$. Multiplying the numerator of ρ by $\bar{c} - k$ gives

$$H(k) = hr - p\bar{c} + (\bar{c} - c_2)k - k^2$$

Here $H(t_P) = 0$ and $H'(t_P) = \bar{c} - c_2 - 2t_P > 0$. Thus for k sufficiently close to t_P , the protected characterization holds with $0 < \rho < 1$. These equilibria approach the maximal pool, while their positive high-demand acceptance excludes the limit from the two-demand class.

Under exposure, choose $x < x_E$ approaching x_E , put $t = t_E(x)$ and $s = \Psi_E(x) > 0$, and set

$$\delta = \frac{-s + \sqrt{s^2 + 4\gamma x s(h-x)}}{2\gamma x}, \quad b = t - \delta, \quad \rho = \frac{\gamma x}{b + c_2}, \quad k = t + \frac{\rho\delta}{1-\rho}, \quad y = h$$

The root satisfies $\gamma x \delta^2 = s(h-x-\delta)$ and tends to zero. Since $t_E > \max\{0, t_h\}$ and $t_E < r$, all strict regular incentive restrictions hold sufficiently close to the limit. The target's low-demand surplus is

$$h(r-k) - x(\bar{c} - k + \gamma b) = \Psi_E(x) - (t-b)(k-t) = s - \frac{\rho\delta^2}{1-\rho} = 0$$

The last equality uses $\frac{\rho}{1-\rho} = \gamma \frac{x}{h-x-\delta}$. Thus these are regular equilibria with low-demand indifference. At $\beta = 1$,

$$\bar{c}q = t, \quad K = I(t) + \frac{\rho\delta^2}{2\bar{c}(1-\rho)} \rightarrow I(t_E)$$

Target indifference gives $U_2 = 1 - hF(r)$, so U and V also approach their pooling endpoints. The high-demand group has positive mass along the sequence but vanishes in the limit. Every finite member satisfies the Intuitive Criterion by [Lemma 3](#).

At each sequence member, any $\beta \in (0, 1]$ is admissible. Varying β fills the interval between its pure-low-response outcome and universal resistance for each scalar outcome. Taking sequence members sufficiently close to the pooling limit reaches every interior value in each stated interval.

Finally, pure pooling war and conflict costs decrease as the accepted demand increases, whereas $U(x) = x + \frac{t_j(x)^2}{2\bar{c}}$ has derivative $1 - \frac{(1-g_j)t_j(x)}{\bar{c}} > 0$. Mixed pooling responses combine these outcomes with universal resistance. Thus no characterized pooling equilibrium exceeds the favorable endpoint or the opposite resistance endpoint. Maximal pooling with certain compliance and all-resisted equilibria attain them, respectively. This closes the four marginal ranges and proves the common extremal comparison. $\square$

V.2. Pairwise Reversals with Certain Low-Demand Acceptance

The nested marginal ranges do not rank arbitrary equilibrium pairs. At every admissible parameter vector, one game's equilibria can approach its favorable endpoint while the other's approach universal resistance. This gives both directions of comparison for q, K, U, V .

Reversals also occur with $\beta = 1$ and strict low-demand acceptance incentives. Set $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.30$, and $\bar{c} = 1.50$, with $y = h = 0.62$ in all four profiles. Consider protected profiles with $(x, k, \rho) = (0.01, 0.60, \frac{1}{62})$ and $(0.30, 0.32, \frac{1}{17})$. Their unnormalized low-demand surpluses are 0.177 and 0.0056. The first exposed profile is the interior example in [Supporting Information P](#). The second has $x = 0.01$, $\rho = \frac{3}{220}$, $b = 0.20$, and $k = \frac{6493}{10850}$, also with strictly positive low-demand surplus. All four satisfy their characterization conditions.

Comparing the interior exposed profile with the first protected profile gives

$$\begin{aligned} \Delta P_C &= \frac{212}{775} > 0, \quad \Delta T = \frac{2119}{15000} > 0, \quad \Delta q = -\frac{202}{2325} < 0 \\ \Delta K &= -\frac{43309}{930000} < 0, \quad \Delta U = \frac{135007}{930000} > 0, \quad \Delta V = \frac{43309}{930000} > 0 \end{aligned}$$

Comparing the second exposed profile with the second protected profile reverses all six signs, and the sign of $\Delta\Omega$ reverses as well. Strict inequalities and continuity preserve these comparisons

in neighborhoods of the primitives and the independently chosen menus. Thus the reversals do not require low-demand compliance to approach zero.

The joint outcomes also distinguish the games. Under protection, compliance and war are disjoint, so $P_C + q \leq 1$. In regular exposed sorting with $\beta = 1$,

$$P_C + q = 1 + \frac{b}{c} > 1$$

No equilibrium in the protected compliance game can reproduce that pair of compliance and war probabilities. The marginal nesting of q, K, U, V therefore does not imply replication of joint outcomes.

W. DISTRIBUTION, INFORMATION, AND ZERO-EXPOSURE BENCHMARKS

The results in this section replace uniformity by an atomless, strictly increasing CDF F with full support on $[0, \bar{c}]$. The remaining maintained assumptions continue to hold, except that the final benchmark sets $\gamma = 0$. Write $I(z) = \int_0^z (c + c_2) dF(c)$. These results concern regular sorting and maximal pooling at zero exposure; they do not extend the uniform-prior outcome bounds in [Proposition 6](#).

W.1. Regular Sorting under a General Prior

We first examine whether feigned restraint and the outcome comparisons with protected compliance depend on a uniform distribution of war costs. The result below allows any prior in the class just specified, with demand sizes and acceptance probabilities adjusting to satisfy the target's willingness to comply.

LEMMA 4 (Regular feigned restraint and its matched comparisons survive a general prior): A regular exposed profile has $y = h$, $0 < x < h$, $0 < \rho < 1$, and

$$b = \frac{\gamma x}{\rho} - c_2, \quad k = \frac{p - x + \rho c_2}{1 - \rho}, \quad \max\{0, t_h\} < b < t_x < k < r$$

Its remaining acceptance constraint is

$$x[1 - F(k) + \gamma F(b)] \leq h[F(r) - F(k)]$$

These conditions are necessary and sufficient, with $\beta = 1$ at strict inequality and any $\beta \in (0, 1]$ at equality. Such profiles exist for every admissible $\gamma > 0$. At $\beta = 1$, the same demands and responses support a protected monotone counterpart. Both profiles satisfy the Intuitive Criterion, and their outcome differences are

$$\Delta P_C = (1 - \rho)F(b) > 0, \quad \Delta T = -\rho h F(b) < 0$$

$$\Delta q = \rho F(b) > 0, \quad \Delta K = \rho I(b) > 0$$

$$\Delta U = \rho \int_0^b (b - c) dF(c) > 0, \quad \Delta U_2 = -\rho(b + c_2)F(b) < 0, \quad \Delta V = -\rho I(b) < 0$$

Peaceful concessions per compliance, Ω , also fall.

Proof. Demand incentives are pointwise in the cost. The ratio $\frac{G_E(x,c)}{G_E(h,c)}$ is constant, decreasing, increasing, and constant on the branches displayed in the feigned-restraint characterization. Its two strict crossings with ρ determine the same b and k under every prior. The high-demand group consists of resolved settlers. Since the target mixes there, its indifference requires $y = h$. In the low-demand group, attackers below b generate an acceptance loss γx , resolved settlers between k and r generate a gain $h - x$, and weak settlers above r generate a loss x . The unnormalized acceptance surplus is therefore

$$S_F = h[F(r) - F(k)] - x[1 - F(k) + \gamma F(b)]$$

This proves the characterization and the response restrictions.

For existence, choose a fixed $b_0 \in (\max\{0, \gamma h - c_2\}, p)$ and set

$$\rho(x) = \frac{\gamma x}{b_0 + c_2}, \quad k(x) = \frac{p - x + \rho(x)c_2}{1 - \rho(x)}$$

As x tends to zero from above, $\rho(x)$ tends to zero, t_x and $k(x)$ tend to p , and

$$k(x) - t_x = \frac{\rho(x)(t_x - b_0)}{1 - \rho(x)} > 0$$

for all sufficiently small positive x . All strict cutoff restrictions then hold. Continuity and full support give

$$S_F \rightarrow h[F(r) - F(p)] > 0$$

so the low demand is surely accepted. No density or hazard-rate restriction is required.

Under protection, the same h, x, ρ have a single sender crossing at k . The low-demand acceptance surplus becomes $S_F + \gamma x F(b) > 0$, while high-demand indifference is unchanged. This proves the matched counterpart at $\beta = 1$. An exposed profile with $\beta < 1$ cannot be matched at the same responses, because the protected target strictly prefers low-demand compliance.

The off-path construction in [Lemma 3](#) does not use uniformity: sender equilibrium dominance is pointwise, and the on-path posterior is the same F conditioned on its demand cells. In particular, weak types share the low-demand payoff $w = \beta x - (1 - \beta)a$, and all types can imitate it. For an unused $d \in [0, h]$ with $d \geq w$, weak types survive the dominance test; a posterior on surviving weak types and resistance deters deviation. If $d < w$, settlement yields less than the imitable payoff. An attacking type also gains less from d than from the used low demand because $d < w \leq \beta x$ and $\gamma > 0$. Thus no response creates a profitable deviation there. A posterior on surviving types and a receiver best response completes this case. For $d > h$, resistance is strictly optimal under every posterior. The exceptional zero-cost protected survivor when $c_2 = 0$ is treated as in [Lemma 3](#). Both profiles therefore satisfy the same refinement under the changed prior.

Only types below b change behavior in the matched pair. Under protection, they demand h , settle with probability ρ , and fight otherwise. Under exposure, they obtain x certainly and attack. Their changes in compliance, peaceful transfer, and war probability are $1 - \rho$, $-\rho h$, and ρ . Their payoff change is

$$p + \gamma x - c - [p - c + \rho(c + c_2)] = \rho(b - c)$$

using $\gamma x = \rho(b + c_2)$. The target loses γx for each such type. Integrating proves every displayed identity. Since $T_P > 0$, $\Delta T < 0$, and $\Delta P_C > 0$,

$$\Delta \Omega = \frac{\Delta T - \Omega_P \Delta P_C}{P_{C,E}} < 0$$

□

The prior can change which menus satisfy the target's constraint. For example, at $p = 0.60$, $c_2 = 0.02$, $a = 0.30$, $\gamma = 0.30$, and $\bar{c} = 1.50$, replacing uniformity by $F(c) = \left(\frac{c}{1.50}\right)^2$ makes the interior menu $x = \rho = 0.20$ infeasible: $S_F = -\frac{39743}{1500000} < 0$. The menu $x = \rho = 0.10$ instead satisfies $S_F = \frac{114481}{2278125} > 0$. Existence under every prior therefore allows demands and responses to adjust. The uniform restriction $x < p$ is also unnecessary for general-prior regular sorting; the lemma uses $x < h$ together with the cutoff restrictions.

W.2. What Demand Signals Reveal

We next compare what the two demand signals reveal about whether the coercer would fight if refused. This separates information about that binary decision from information about the coercer's exact war cost.

LEMMA 5 (Exposure reduces information about binary resistance resolve): Fix F and a matched regular pair from Lemma 4. For $Z = \mathbb{1}[c < r]$, the exposed demand signal is a strict Blackwell garbling of the protected signal. Relabel a protected high demand as low with probability $\frac{F(b)}{F(k)}$ and leave a protected low demand unchanged. The two demand experiments are Blackwell incomparable about the full cost c .

Proof. Use the comparison of experiments in Blackwell (1951). Let $A = F(k)$, $B = F(b)$, and $\pi = F(r)$, so $0 < B < A < \pi < 1$. Conditional on $Z = 0$, both games always produce the low demand. Conditional on $Z = 1$, their high-demand probabilities are $\frac{A}{\pi}$ and $\frac{A-B}{\pi}$. The stated kernel reduces the first probability to the second without conditioning on the realized binary state. A reverse kernel must keep an exposed low demand low, because it is the only signal in state $Z = 0$. It can therefore produce a high demand in state $Z = 1$ with probability at most $\frac{A-B}{\pi} < \frac{A}{\pi}$. The binary order is strict.

For the full cost, protection partitions the support into $[0, k)$ and $[k, \bar{c}]$, while exposure partitions it into $[b, k)$ and $[0, b) \cup [k, \bar{c}]$. A garbling of the protected high signal cannot send it to low for $c < b$ and to high for $b < c < k$. A reverse garbling of the exposed low signal cannot send it to high for $c < b$ and to low for $c > k$. All three intervals have positive prior

mass, so neither garbling exists. Equivalently, protection identifies $\mathbb{1}[c < k]$ perfectly, whereas exposure identifies $\mathbb{1}[b < c < k]$ perfectly; each classification has positive error under the other experiment. $\square$

The binary comparison averages over the fixed prior's costs within each resolve state. Its kernel depends on $\frac{F(b)}{F(k)}$ and is not a common kernel for arbitrary changes in that within-state distribution. Nor does the result rank information about willingness to stop after compliance. That decision depends on the demand and attack technology. The target's actual payoffs also depend on these features, so binary informativeness alone does not sign its equilibrium payoff.

W.3. Retaining the Attack Option at Zero Exposure

To isolate the military effect of concessions, we now remove that effect while retaining the option to attack after compliance. We compare the outcomes at each game's largest pooling demand when the target accepts with the same probability.

LEMMA 6 (Zero exposure preserves maximal-pooling material outcomes): Set $\gamma = 0$ while retaining the exposed game's attack option. Let $t(x) = \max\{p - x, 0\}$ and

$$\Psi_F(x) = h[F(r) - F(t(x))] - x[1 - F(t(x))]$$

Both games have the same maximal pooling demand

$$x^* = \max\{x \in [0, 1] : \Psi_F(x) \geq 0\} \in (0, h), \quad t^* = t(x^*)$$

with $\Psi_F(x^*) = 0$. Every common compliance rate $\lambda \in (0, 1]$ is supportable at this demand. Terminal settlement, war, and backing-down probabilities and payoffs agree type by type almost surely. Consequently $(T, q, K, U, U_2, V)_E = (T, q, K, U, U_2, V)_P$.

In the exposed game, any prior mass $m \in [0, F(t^*)]$ of attacking types may join the accepted pool. Thus

$$P_{C,P} = \lambda[1 - F(t^*)], \quad P_{C,E} = \lambda[1 - F(t^*) + m]$$

$$\Omega_P = x^*, \quad \Omega_E = x^* \frac{1 - F(t^*)}{1 - F(t^*) + m}$$

A strict gap in Ω occurs exactly when $m > 0$. Regular feigned restraint with strict demand preferences disappears at $\gamma = 0$.

Proof. Under protection, types above $t(x)$ strictly prefer an accepted positive demand to resistance, and types below it prefer resistance. Under exposure at $\gamma = 0$, the former still join the accepted pool. The latter earn $p - c$ from either acceptance followed by attack or resistance followed by war. They are indifferent about joining. Full support and atomlessness permit any accepted-attacker mass between zero and $F(t(x))$.

An accepted attacker gives the target identical compliance and resistance payoffs. The remaining accepted types generate total acceptance surplus $\Psi_F(x)$ in either game. The different posterior denominators do not change its sign. Thus $\Psi_F(x) \geq 0$ characterizes demands feasible at some positive compliance rate. Strict inequality requires $\lambda = 1$; equality permits every positive $\lambda \leq 1$. Assign excluded attackers to demand 1, which is surely resisted because $h < 1$. On-path incentives hold as stated. The completion in [Lemma 3](#) extends to zero exposure by retaining attacking types that are indifferent in its survivor set: for $d < w$, attacking at an unused demand yields $R(c) \leq v(c)$, so no deviation is profitable even when dominance is weak. The remaining off-path cases are unchanged.

Continuity gives $\Psi_F(0) = h[F(r) - F(p)] > 0$ and $\Psi_F(h) = h[F(r) - 1] < 0$. No $x \geq h$ is feasible. The largest feasible demand therefore exists in $(0, h)$ and binds the target's constraint. This proves common maximality and independence from λ without requiring a unique positive root.

At x^* , types below t^* fight surely in both games. Above t^* , the common acceptance rate gives the same settlement, war, and backing-down lotteries. Since the concession is not retained separately after war, changing the history from resistance and war to compliance and attack changes no material payoff at $\gamma = 0$. This proves typewise equivalence and the aggregate equality. Counting accepted attackers adds λm to compliance without changing the peaceful transfer $\lambda x^*[1 - F(t^*)]$, which gives the formulas for Ω .

Finally, in regular feigned restraint the high-demand pool consists of resolved settlers, so target mixing requires $y = h$. A positive-cost low-demand attacker gains zero from compliance with x but gains $c + c_2 > 0$ from compliance with h . Positive high-demand compliance therefore rules out its strict preference for the low demand. The cutoff equation $\rho(b + c_2) = \gamma x = 0$ cannot hold with $\rho, b > 0$. $\square$

Under uniformity and Assumption 1, $x^* = x_P < p$ and $t^* > 0$. The all-type exposed pool sets $m = F(t^*)$ and has $\Omega_E = x^*[1 - F(t^*)] < \Omega_P$, despite equal material outcomes. It is one admissible composition; setting $m = 0$ makes the compliance counts coincide too. For a general prior, if $hF(r) \geq p$, the common maximal demand is $x^* = hF(r)$ and $t^* = 0$, so no strict composition effect is possible. If $hF(r) < p$, then $x^* < p$. These alternatives follow from $\Psi_F(x) = hF(r) - x$ for $x \geq p$.

The zero-exposure result does not equate complete equilibrium correspondences. Types that would attack after either demand can be indifferent between multiple positive demands. Such assignments can survive at $\gamma = 0$ even though the strict incentives supporting regular feigned restraint disappear. The statement also excludes $\lambda = 0$, at which coercive ability conditional on compliance is undefined.

X. ACCEPTED DEMAND COUNTS ACROSS THE TWO GAMES

This section establishes the protected-compliance part of [Proposition 2](#). The bound follows from sender and target optimality; it does not require an additional equilibrium refinement.

LEMMA 7 (Accepted demands under protected compliance): In any PBE in the protected compliance game with pure demand strategies, at most two distinct strictly positive demands can each be chosen by a positive prior mass of types and accepted with positive probability. If there are two, the larger equals $h = p + c_2$ and is chosen only by resolved types, up to sets of prior probability zero. The bound permits other demands that are zero or always resisted.

Proof. Take any two distinct positive accepted demands $x < y$, and write $\lambda_x = \lambda(x)$ and $\lambda_y = \lambda(y)$. A type choosing x must have $G_{P(x,c)} \geq 0$: otherwise it could demand 1, induce resistance, and obtain a strictly higher payoff. Since $G_{P(y,c)} = G_{P(x,c)} + y - x$, the inequality $\lambda_y \geq \lambda_x$ would make y strictly preferable for every type choosing x . Hence $0 < \lambda_y < \lambda_x \leq 1$.

Weak types must strictly prefer x to y . To prove this, suppose instead that $\lambda_{y(y+a)} \geq \lambda_{x(x+a)}$. For every resolved type $c < r$, the difference between the gains from the two demands is

$$\begin{aligned} \lambda_y G_{P(y,c)} - \lambda_x G_{P(x,c)} &= \lambda_{y(y+a)} - \lambda_{x(x+a)} \\ &\quad + (\lambda_y - \lambda_x)(c - r) > 0. \end{aligned}$$

Every resolved type would therefore strictly prefer y . The pool choosing x could contain only weak types, against whom the target strictly prefers resistance to conceding $x > 0$. This contradicts positive compliance. Thus weak types strictly prefer the smaller of any two positive accepted demands.

Now suppose three such demands $x < y < z$ existed. The preceding argument excludes weak types from both y and z and gives $0 < \lambda_z < \lambda_y < \lambda_x \leq 1$. The target must therefore mix after both higher demands. Because their coercers are resolved, resistance gives the target $1 - h$, whereas accepting a demand d gives it $1 - d$. Indifference requires $y = h = z$, contradicting distinct demands. The same reasoning implies that the larger of exactly two positive accepted demands equals h and contains only resolved types. Zero and always-resisted demands do not change this pairwise argument. $\square$

Every equilibrium satisfying the Intuitive Criterion is a PBE, so the bound also applies under the paper's refinement. It rules out three or more distinct positive accepted demands under protected compliance. The next section proves the exposed-compliance part of [Proposition 2](#), including the existence threshold and the pattern in which only the middle demand carries attack risk. It also extends the outcome bounds in [Proposition 6](#) to finite positive accepted-demand sets, allowing comparisons across different demand counts.

Y. THREE-DEMAND EQUILIBRIA UNDER EXPOSED COMPLIANCE

This section characterizes equilibria in the exposed compliance game with deterministic demand choices whose entire set of on-path demands consists of exactly three strictly positive amounts. Each demand is strictly positive, is chosen by a positive prior mass of types, and receives positive compliance. The prior remains uniform, and Assumptions 1–2 are unchanged. An additional used zero demand is outside this class. Every type strictly prefers a positive demand receiving positive compliance to a surely resisted demand, so no additional refused demand can attract positive mass.

The three-demand equilibria in [Proposition 2](#) have three forms: attackers strictly prefer the middle demand, are indifferent between the low and middle demands, or are indifferent between the middle and high demands. Weak coercers choose the lowest demand, and acceptance becomes less likely as the demand rises. The conditions below characterize every form, including all admissible target responses and assignments of indifferent coercers. Every admissible on-path profile has a completion satisfying the Intuitive Criterion.

The existence condition is

$$t_h = \gamma h - c_2 > 0.$$

By [Proposition 1](#), this is precisely the region in which every equilibrium with two positive accepted demands involves a risk of attack after compliance. Three-demand bargaining therefore becomes possible above the same exposure threshold.

Order the demands and write their acceptance probabilities as

$$0 < x < z < y \leq h, \quad (\lambda_x, \lambda_z, \lambda_y) = (\beta, \beta\sigma, \beta\sigma\rho),$$

$$0 < \beta \leq 1, \quad 0 < \sigma < 1, \quad 0 < \rho < 1.$$

Thus σ is the middle demand's response relative to the low demand's response, and ρ is the high demand's response relative to the middle demand's. Write $t_d = p - (1 - \gamma)d$. The quantities describing sets of attackers below are lengths in the cost support; dividing by $\bar{c}$ gives their prior probabilities.

For each family, let S_x be the target's low-demand compliance surplus multiplied by its positive posterior denominator. The response restriction is

$$S_x \geq 0, \quad (1 - \beta)S_x = 0.$$

A strict surplus requires certain low-demand compliance. At equality, every $\beta \in (0, 1]$ is admissible, with both higher responses scaled in the same proportion. Both higher responses remain strictly between zero and one. The following conditions apply when $t_h > 0$ and disregard assignments of cutoff types with prior probability zero.

Y.1. Attackers Strictly Prefer the Middle Demand

Set $y = h$, and define

$$\rho_{\min} = \frac{\gamma z}{h - (1 - \gamma)z}, \quad \rho_{\min} < \rho < \frac{z}{h},$$

$$b = \frac{\gamma z}{\rho} - c_2, \quad k = \frac{p - z + \rho c_2}{1 - \rho}, \quad \ell = k + \frac{\gamma z b}{h - z}.$$

The remaining conditions are

$$p < \ell < r, \quad \sigma = \frac{x + \ell - p}{z + \ell - p}, \quad S_x = h(r - \ell) - x(\bar{c} - \ell) \geq 0.$$

Together with $0 < x < z < h$, these restrictions imply

$$0 < t_h < b < t_z < k < p < \ell < r, \quad \sigma z > \max\{x, \sigma \rho h\}.$$

The demand assignment is

$$d(c) = \begin{cases} z & \text{if } 0 \leq c < b \\ h & \text{if } b \leq c < k \\ z & \text{if } k \leq c < \ell \\ x & \text{if } \ell \leq c \leq \bar{c}. \end{cases}$$

Types below b attack after compliance. Every other type settles after compliance, and each type strictly prefers its assigned demand except at b , k , and ℓ . The middle demand attracts both the lowest-cost attackers and an interval of resolved settlers. Weak types choose the lowest demand.

Y.2. Attackers Are Indifferent between the Low and Middle Demands

Set $y = h$, require $\rho_{\min} < \rho < \frac{z}{h}$, and define b and k as in the preceding family. Now impose

$$\sigma = \frac{x}{z}, \quad m_z = \frac{(h-z)(p-k)}{\gamma z}, \quad 0 < m_z \leq b, \quad m_x = b - m_z.$$

The low-demand constraint becomes

$$S_x = ha - x(\bar{c} - p + \gamma m_x) \geq 0.$$

The cutoffs satisfy $0 < t_h < b < t_z < k < p < r$. Every type in $[0, b)$ is indifferent between x and z and attacks after either demand is accepted. Choose any measurable subset of $[0, b)$ with total length m_x to demand x ; its complement demands z . Types in $[b, k)$ demand h , types in $[k, p)$ demand z , and types in $[p, \bar{c}]$ demand x . All types in these three upper groups settle after compliance.

A canonical representation assigns $[0, m_x)$ to the low demand and $[m_x, b)$ to the middle demand. Arbitrary measurable assignments with the same lengths also work. The boundary $m_x = 0$ is included: all attackers then choose z , although they remain indifferent between the two lower demands. The middle demand must receive a positive mass of attackers. Here

$$x = \sigma z > \sigma \rho h.$$

Y.3. Attackers Are Indifferent between the Middle and High Demands

Allow $y \leq h$ and impose

$$\rho = \frac{z}{y}, \quad m_y = \frac{(h-y)(p-t_y)}{\gamma y} = \frac{(1-\gamma)(h-y)}{\gamma},$$

$$m_z = t_y - m_y = t_h - \frac{(1-\gamma)^2(h-y)}{\gamma} > 0.$$

The nonnegative length m_y is the high-demand attacker group, and m_z is the middle-demand attacker group. These restrictions imply $0 < t_y < p$. Define

$$\ell = p + \frac{\gamma z m_z}{h-z}, \quad p < \ell < r, \quad \sigma = \frac{x + \ell - p}{z + \ell - p}.$$

The low-demand constraint is

$$S_x = h(r - \ell) - x(\bar{c} - \ell) \geq 0.$$

Every type in $[0, t_y)$ is indifferent between z and y and attacks after either is accepted. Choose any measurable subset of this interval with total length m_y to demand y ; its complement demands z . Types in $[t_y, p)$ demand y , types in $[p, \ell)$ demand z , and types in $[\ell, \bar{c}]$ demand x . All types in these upper groups settle after compliance.

A canonical representation assigns $[0, m_y)$ to the high demand and $[m_y, t_y)$ to the middle demand. The boundary $y = h$ gives $m_y = 0$, so only the middle demand receives attackers. When $y < h$, both tied demands receive a positive mass of attackers. In this family,

$$\sigma z = \sigma \rho y > x.$$

Y.4. Proof of the three-demand claims in [Proposition 2](#)

Proof. The protected impossibility in [Proposition 2](#) follows from [Lemma 7](#). For exposure, we establish necessity from target optimality and the sender's gain envelope, verify sufficiency and the Intuitive Criterion, and then prove the exact existence region. The strict first family establishes the additional signaling pattern: the middle demand can carry attack risk while both extreme demands assure settlement. The extension to the full finite class follows in [Supporting Information Y.5](#).

Demand and response restrictions. Against a resolved settler, a demand $d > h$ costs the target more than resistance. Against a weak type it costs more than backing down, and against an attacker it adds $\gamma d > 0$ to the resistance loss. Thus every posterior induces resistance at $d > h$, and the highest accepted demand satisfies $y \leq h$. The exposed compliance gain is strictly positive and strictly increasing in the demand for every type. If a larger used demand received compliance at least as often as a smaller one, every type assigned to the smaller demand would strictly prefer to mimic the larger one. Hence

$$0 < \lambda_y < \lambda_z < \lambda_x \leq 1.$$

Weak types strictly prefer the smaller of any two positive accepted demands. To see this, suppose $d_1 < d_2$ but $\lambda_2(d_2 + a) \geq \lambda_1(d_1 + a)$. For a resolved type that settles after compliance with d_1 ,

$$\begin{aligned} \lambda_2 G_E(d_2, c) - \lambda_1 G_E(d_1, c) &\geq \lambda_2(d_2 + a) - \lambda_1(d_1 + a) \\ &\quad + (\lambda_2 - \lambda_1)(c - r) > 0. \end{aligned}$$

No resolved settler would choose d_1 . Its pool could contain only weak types and attackers, both of whom impose a strictly negative compliance surplus at a positive demand. This contradicts positive compliance. Therefore all weak types choose x , and both higher pools contain only resolved types. Both higher target responses are interior. Since $z < y \leq h$, target indifference at z requires a positive mass of attackers and a positive mass of resolved settlers.

The sender's gain envelope. For a resolved type, divide the gain from the three available demands by β . Let

$$M = \max\{x, \sigma z, \sigma \rho y\}.$$

The best available gain is

$$H(c) = \max\{\gamma M, x - p + c, \sigma(z - p + c), \sigma \rho(y - p + c)\}.$$

A type that attacks at its chosen demand must use a demand whose probability-times-demand product equals M . Because the middle pool contains attackers,

$$\sigma z = M \geq \max\{x, \sigma \rho y\}.$$

The three settlement lines have distinct positive slopes. Their intersections with one another or with the constant branch occur at isolated costs. Positive-mass indifference among resolved types can therefore arise only when two maximal constant branches coincide.

The high product is strictly smaller. Suppose $\sigma \rho y < \sigma z$. No attacker chooses y , so target mixing requires $y = h$. The high settlement line crosses the maximal constant at b and the middle settlement line at k . A positive high-demand interval requires

$$b < k \equiv \rho > \frac{\gamma z}{h - (1 - \gamma)z}.$$

The strict product inequality gives $\rho < \frac{z}{h}$, and these inequalities imply $b < t_z < k < p$. The positive middle-attacker mass also requires $b > 0$.

If $x < \sigma z$, the middle and low settlement lines cross at a unique $\ell > p$, satisfying

$$\ell = p + \frac{\sigma z - x}{1 - \sigma}.$$

Weak types' strict preference for x requires $\ell < r$. The envelope assigns z below b , h between b and k , z between k and ℓ , and x above ℓ . This is the first family's assignment. If $x = \sigma z$, the middle and low settlement lines instead cross at p . Both demands share the maximal constant below b , so the attackers there can be divided between them. This gives the second family's assignment.

The high and middle products are equal. Suppose $\sigma \rho y = \sigma z$. Then $\rho = \frac{z}{y}$. If $x < \sigma z$, the high settlement line dominates the middle line below p , while the middle dominates above p until the low line takes over at $\ell > p$. The constant ends at t_y . The middle pool's required attackers imply $t_y > 0$, and weak types' strict preference again gives $\ell < r$. These are precisely the third family's assignment sets.

If all three products are equal, the high settlement line dominates the middle line for $c < p$ and the low line dominates it for $c > p$. The middle demand could contain only attackers, apart from the probability-zero type $c = p$. Its target response would then be resistance. Triple equality is impossible. These cases exhaust all product orders. Ties between nonmaximal products create no further assignments because their constant branches are never optimal, and the weak-type ranking already excludes upper-plateau ties.

Target optimality and the permitted attacker assignments. Uniformity makes the target's surplus depend on the lengths of the assigned attacker and settler sets, including when a tied attacker set is not an interval. In the first family, the high-demand surplus is zero and middle-demand indifference requires

$$(h - z)(\ell - k) - \gamma z b = 0.$$

This determines ℓ . In the second family, middle-demand indifference requires

$$(h - z)(p - k) - \gamma z m_z = 0,$$

which determines the required middle-attacker length and leaves $m_x = b - m_z$. In the third family, high- and middle-demand indifference require

$$\begin{aligned}(h - y)(p - t_y) - \gamma y m_y &= 0, \\ (h - z)(\ell - p) - \gamma z m_z &= 0,\end{aligned}$$

together with $m_y + m_z = t_y$. These equations give exactly the lengths and cutoff stated above.

The resulting low-demand surplus is $h(r - \ell) - x(\bar{c} - \ell)$ in the first and third families. In the second family it is

$$(h - x)(r - p) - x(\bar{c} - r) - \gamma x m_x = ha - x(\bar{c} - p + \gamma m_x).$$

Thus target optimality is exactly $S_x \geq 0$ and $(1 - \beta)S_x = 0$. The relative responses and all sender comparisons are unchanged by the common scale β .

Conversely, each family's conditions give the stated envelope order, positive mass at every demand, and optimal target responses under Bayes' rule. The arbitrary measurable attacker assignments preserve both sender indifference and the required target surplus. Each type strictly exceeds its resistance payoff because some positive demand receives positive compliance. Weak types have the common low-demand payoff

$$w = \beta x - (1 - \beta)a.$$

Apply [Lemma 3](#) with $s = x$ and $\lambda_s = \beta$ to complete every such profile. Its beliefs deter all unused demands without changing on-path behavior, including at target-mixing boundaries and with noninterval attacker sets. Necessity holds for every PBE, and sufficiency supplies a completion satisfying the Intuitive Criterion.

The existence region. It remains to establish why all three families require $t_h > 0$ and why this condition suffices. For the first two families, define

$$D(\rho) = k - p + \frac{\gamma z b}{h - z} = -\frac{z - \rho h}{1 - \rho} + \frac{\gamma z (\gamma_{\rho}^{\frac{z}{\rho}} - c_2)}{h - z}.$$

On $\rho_{\min} < \rho < \frac{z}{h}$,

$$D'(\rho) = \frac{h - z}{(1 - \rho)^2} - \frac{\gamma^2 z^2}{\rho^2 (h - z)} > 0.$$

All denominators are positive. The derivative inequality is equivalent to $\rho > \rho_{\min}$. As ρ approaches the open upper endpoint from below,

$$\lim_{\rho \rightarrow \frac{z}{h}} D(\rho) = \frac{\gamma z t_h}{h - z}.$$

The first family requires $D = \ell - p > 0$. The second requires

$$D = \frac{\gamma z (b - m_z)}{h - z} \geq 0.$$

Strict monotonicity and the open upper endpoint therefore imply $t_h > 0$ in both families, including the second family's $m_x = 0$ boundary. In the third family,

$$0 < m_z = t_h - \frac{(1 - \gamma)^2 (h - y)}{\gamma} \leq t_h,$$

so the same condition is necessary.

For sufficiency, suppose $t_h > 0$. Choose $z > 0$ small enough that $z < h$ and $\gamma z \frac{t_h}{h - z} < a$. Choose $\rho < \frac{z}{h}$ sufficiently close to $\frac{z}{h}$, while exceeding $\rho_{\min}$. Then $0 < D < a$, so the first family's cutoffs satisfy $p < \ell < r$. Choose

$$0 < x < \min \left\{ z, \frac{h(r - \ell)}{\bar{c} - \ell} \right\}, \quad \sigma = \frac{x + \ell - p}{z + \ell - p}, \quad \beta = 1.$$

All first-family conditions hold, with strict low-demand acceptance and strict sender preferences away from the cutoffs. The completion above gives an equilibrium satisfying the Intuitive Criterion.

Both tie families are nonempty in the same region. For the second, take ρ close enough to $\frac{z}{h}$ from below that $D > 0$, yielding $0 < m_z < b$, and choose $x > 0$ sufficiently small for strict low acceptance. For the third, choose $y < h$ sufficiently close to h that $m_z > 0$, then choose $z > 0$ small enough that $\ell < r$ and $x > 0$ small enough for strict low acceptance. Both tied attacker groups can have positive mass. No three-demand equilibrium exists at $t_h = 0$. This proves the characterization and the exact existence condition. $\square$

The characterization includes certain low-demand compliance at indifference, all positive mixed low-demand responses allowed by the surplus condition, the zero low-attacker boundary

in the second family, the zero high-attacker boundary in the third, and zero target war costs. Zero demands, zero compliance rates, coincident amounts, and zero mass at a used demand remain outside the stated class. The middle demand need not lie below p ; the low-demand acceptance constraint does imply $x < p$. The result characterizes three used demands and does not assert that three is the largest number the exposed game can sustain.

Y.5. Higher Demand Counts in the Existence Map

Under the maintained assumptions, uniform prior, and deterministic demand choices, an equilibrium with a finite set of three or more strictly positive accepted demands, each chosen by positive prior mass, exists under exposure if and only if $\gamma h > c_2$. No such equilibrium exists under protection. Thus the gray region in panel (b) of [Figure 2](#) is exactly the region above the curve; the curve itself is excluded. The accepted-demand set in this statement contains no zero demand.

Proof. The protected impossibility follows from [Lemma 7](#) in [Supporting Information X](#). Exposed sufficiency follows from the three-demand construction in [Supporting Information Y.4](#). For necessity, we first bound the number of positive accepted demands. Acceptance probabilities strictly decrease with demand, and weak coercers strictly prefer the lowest demand. Each higher accepted demand below h must therefore contain both attackers and resolved settlers to make the target indifferent. Because it attracts attackers, its acceptance-weighted amount must maximize $\lambda_d d$. Denote this maximum by M . The resolved settlement gain at any such demand is

$$M + \lambda_{d(c-p)}.$$

Among demands sharing M , only the largest can maximize this expression below p , and only the smallest can maximize it above p . A demand strictly between them cannot attract a positive mass of settlers. Thus at most two demands above the lowest can lie below h . Including the lowest and possibly h gives at most four positive accepted demands.

Necessity for exactly three demands is established in [Supporting Information Y.4](#). If four demands are used, they must be $x < z < y < h$, with h the fourth amount. The middle products

satisfy $\lambda_z z = \lambda_y y = M$. Both endpoint products must be strictly smaller; equality at either endpoint would deprive one middle demand of its required settler group. Write $\rho = \frac{\lambda_h}{\lambda_y}$. The highest demand attracts settlers on $[b, k)$, the upper middle demand attracts settlers on $[k, p)$, and the lower middle demand attracts settlers above p , where

$$b = \frac{\gamma y}{\rho} - c_2, \quad k = \frac{p - y + \rho c_2}{1 - \rho}, \quad \frac{\gamma y}{h - (1 - \gamma)y} < \rho < \frac{y}{h}.$$

The two middle demands divide all attackers in $[0, b)$. Let their positive lengths be m_z and m_y , so $m_z + m_y = b$. Target indifference after y requires $(h - y)(p - k) = \gamma y m_y$. Consequently,

$$k - p + \frac{\gamma y b}{h - y} = \frac{\gamma y m_z}{h - y} > 0.$$

The left side is the function $D(\rho)$ in the preceding existence proof, with y replacing z . It is strictly increasing on the displayed open interval and approaches $\gamma \frac{y(h - c_2)}{h - y}$ at its upper endpoint. Positivity therefore requires $\gamma h > c_2$. Four demands cannot occur on or below the curve, and five or more are impossible. The existing three-demand construction proves existence above it, completing the claim for the finite positive accepted-demand class. $\square$

Y.6. Outcomes and Comparisons across Demand Counts

The three families differ in which demands belligerent coercers choose, but their war probabilities have the same form. Use $b = t_y$ and $k = p$ in the family with indifference between the middle and high demands, and $\ell = p$ in the family with indifference between the low and middle demands. Write m_x , m_z , and m_y for the lengths of the attacker sets assigned to the three demands, so $m_x + m_z + m_y = b$. These lengths specify prior probabilities after division by $\bar{c}$.

First set the common acceptance factor to $\beta = 1$. Types below b fight for certain. Resolved settlers in $[b, k)$ face resistance with probability $1 - \sigma\rho$, those in $[k, \ell)$ with probability $1 - \sigma$, and all types above ℓ settle. Consequently,

$$q(1) = \frac{b + (1 - \sigma\rho)(k - b) + (1 - \sigma)(\ell - k)}{\bar{c}},$$

$$K(1) = I(b) + (1 - \sigma\rho)[I(k) - I(b)] + (1 - \sigma)[I(\ell) - I(k)].$$

The expected peaceful transfer and probability of compliance are

$$T(1) = \frac{\sigma\rho y(k-b) + \sigma z(\ell-k) + x(\bar{c}-\ell)}{\bar{c}},$$

$$\begin{aligned}\bar{c}P_C(1) &= m_x + \sigma m_z + \sigma\rho m_y \\ &\quad + \sigma\rho(k-b) + \sigma(\ell-k) + \bar{c} - \ell.\end{aligned}$$

Attacker assignments with the same lengths give the same outcomes. All these types fight regardless of the target's response, and indifference across their demands equates their expected military gain from compliance. The exact cost identities therefore give

$$U(1) = x + K(1) - c_2 q(1), \quad U_2(1) = 1 - K(1) - U(1), \quad V(1) = 1 - K(1).$$

When the target is indifferent after the low demand, any common factor $\beta \in (0, 1]$ is admissible. Lowering this factor leaves the choices of demands unchanged and increases resistance. For each $Y \in \{q, K, U, U_2, V\}$,

$$Y(\beta) = \beta Y(1) + (1 - \beta)Y_R,$$

where the resistance outcomes are defined in [Supporting Information V](#) and $U_{2,R} = 1 - hF(r)$. Also, $T(\beta) = \beta T(1)$ and $P_C(\beta) = \beta P_C(1)$, so $\Omega(\beta) = \frac{T(1)}{P_C(1)}$. The factor is fixed at one when the target strictly prefers accepting the low demand.

Although there is no equilibrium in the protected compliance game with three positive accepted demands, outcome comparisons need not preserve the number of demands. The following result extends the bounds in [Proposition 6](#) to every finite set of positive accepted demands. It therefore covers all three-demand families without imposing an equilibrium selection rule.

LEMMA 8 (Outcome bounds with additional accepted demands): Under the uniform prior, consider any equilibrium satisfying the Intuitive Criterion with pure demand strategies whose on-path accepted demands form a finite set of at least two strictly positive amounts, each chosen by positive prior mass. In either game, war risk and conflict cost are strictly higher, and the coercer's expected payoff and total welfare strictly lower, than at that game's maximal pooling demand accepted with certainty. Adding these equilibria to pooling and the two-demand class leaves the marginal outcome ranges in [Supporting Information V](#) unchanged.

Proof. The protected case follows from [Lemma 7](#) and the two-demand proof of [Proposition 6](#). The higher demand is h , which gives every resolved type a compliance gain $c + c_2$ and every weak type a strictly positive gain. Thus surely resisted choices have zero prior mass, including when $c = c_2 = 0$ is indifferent, and the existing two-demand characterization applies. For the exposed case, let $x > 0$ be the smallest accepted demand and let β be its acceptance probability. Comparing any two accepted demands as in the necessity argument in [Supporting Information Y.4](#) shows that larger demands have strictly lower acceptance probabilities and weak types strictly prefer x . All types obtain a strictly positive gain over resistance from some positive accepted demand, so no type chooses a surely resisted demand.

First take $\beta = 1$. Every weak type obtains x for certain, and backing down never occurs. Let $v(c)$ be the coercer's equilibrium payoff and $w(c)$ its probability of war. Each demand offers a convex, piecewise affine payoff as a function of the war cost. Their finite maximum is absolutely continuous. Optimal continuation and demand choice give $v'(c) = -w(c)$ almost everywhere. Coincident payoff segments have the same derivative, so arbitrary assignments of indifferent types preserve this identity. Since $v(\bar{c}) = x$, integration under the uniform prior yields

$$\bar{c}q = v(0) - x, \quad U = x + K - c_2q.$$

The zero-cost type can imitate x and attack, implying $v(0) \geq p + \gamma x$. Thus $\bar{c}q \geq t_E(x)$.

Every type can obtain at least x , and the target can resist each observed demand. Therefore $U \geq x$ and $U_2 \geq 1 - hF(r)$. With no backing down, $U + U_2 = 1 - K$, so $x \leq h \frac{r}{\epsilon} < p$. The

rearrangement argument in the proof of [Proposition 6](#) gives $K \geq I(\bar{c}q)$. The inequality is strict: a demand above x has an acceptance probability strictly between zero and one and must contain positive mass of resolved settlers. A pool containing only attackers and weak types would be resisted. Those settlers fight with an intermediate probability, ruling out concentration of all war among the lowest-cost types.

Combining the envelope identity, the target's resistance option, and this strict cost bound gives

$$\begin{aligned} hr &\geq \bar{c}(U + K) \\ &> \bar{c}x + (\bar{c}q)^2 + c_2\bar{c}q \\ &\geq \bar{c}x + t_E(x)^2 + c_2t_E(x). \end{aligned}$$

Hence $\Psi_E(x) > 0$, which implies $x < x_E$. It follows that $q > q_E^*$, $K > K_E^*$, $U < U_E^*$, and $V < V_E^*$. The opposite strict bounds also hold: war occurs only among resolved types, some resolved settlers avoid it, every type can guarantee its resistance payoff, and weak types gain strictly. Thus $q < q_R$, $K < K_R$, $U > U_R$, and $V > V_R$.

If $\beta < 1$, the target is indifferent after every accepted demand. Dividing all acceptance probabilities by β preserves every target best response and multiplies all sender gains relative to resistance by the same positive factor. The normalized profile satisfies the completion conditions in [Lemma 3](#). Each original outcome is $Y(\beta) = \beta Y(1) + (1 - \beta)Y_R$, so the same strict bounds follow. Finally, the two-demand constructions in [Supporting Information V](#) already attain every interior value of each marginal range, while pooling and universal resistance attain its endpoints. Additional finite positive demand sets can neither enlarge these ranges nor remove any existing outcome. This conclusion concerns each outcome separately, not the joint set of attainable outcome vectors. $\square$